



Meeting of the Board
30 September – 2 October 2017
Cairo, Arab Republic of Egypt
Provisional agenda item 14(g)

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11 September 2017

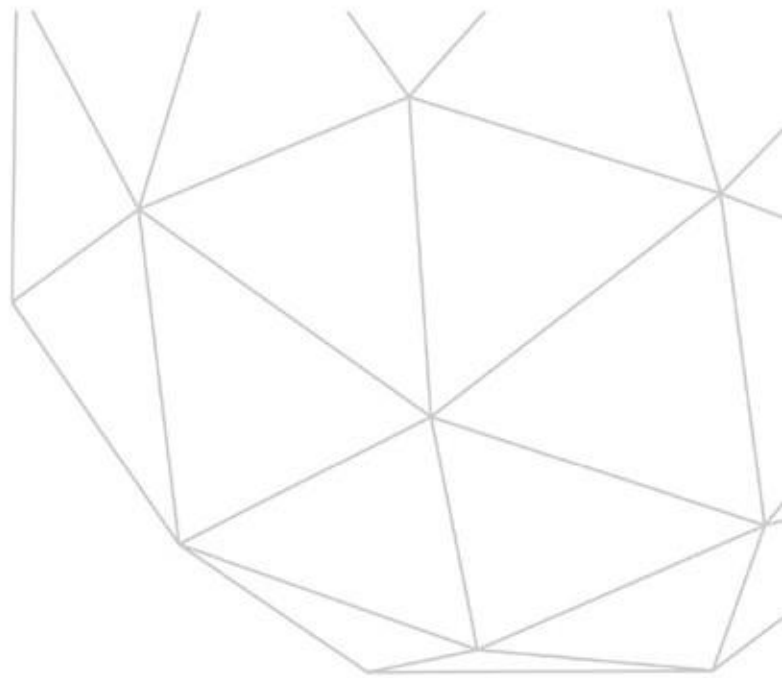
Consideration of funding proposals – Addendum IX

Funding proposal package for FP054

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “Implementation project of the integral management plan of the Lujan River Basin”;
- b) No-objection letters issued by the national designated authority(ies) or focal point(s); and
- c) Environmental and social report(s) disclosure;



Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: Implementation Project for the Management Plan of the Lujan River Basin.

Country/Region: Argentina- Latin America

Accredited Entity: CAF – banco de desarrollo de América Latina

Date of Submission: January 27th, 2016

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Section B	FINANCING / COST INFORMATION
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Section G	RISK ASSESSMENT AND MANAGEMENT
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Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"[FP]-[Agency Short Name]-[Date]-[Serial Number]"

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Implementation Project of the Integral Management Plan of the Lujan River Basin
A.1.2. Project or programme		Project
A.1.3. Country (ies) / region		Argentina
A.1.4. National designated authority (ies)		<i>Subsecretaría de Finanzas del Ministerio de Economía de la Provincia de Buenos Aires (Subsecretariat of Finance of the Ministry of Economy of the Province of Buenos Aires).</i>
A.1.5. Accredited entity		CAF – Banco de Desarrollo de América Latina
A.1.5.a. Access modality		☒ Direct ☐ International
A.1.6. Executing entity / beneficiary		Executing Entity: <i>Province of Buenos Aires through the Ministry of Infrastructure and Public Services and Subsecretary of Finance from Ministry of Economy</i> Beneficiary:
A.1.7. Project size category (Total investment, Million USD)		☐ Micro (≤ 10) ☐ Medium ($50 < x \leq 250$) ☐ Small ($10 < x \leq 50$) ☒ Large (> 250)
A.1.8. Mitigation / adaptation focus		☐ Mitigation ☒ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		27-01-2017
A.1.10. Project contact details	Contact person, position	Antonio Garcia / Martha Castillo
	Organization	Banco de Desarrollo de América Latina - CAF
	Email address	agarcia@caf.com / mcastillo@caf.com
	Telephone number	+511.710.2838 / +571.743.7352
	Mailing address	- Av. Enrique Canaval y Moreyra No 380 Edificio Torre Siglo XXI - Lima - Peru - Carrera 9 No. 76-49, 110221 Bogota - Colombia

A.1.11. Results areas (mark all that apply)	
Reduced emissions from:	
☐	Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)
Increased resilience of:	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)
☒	Infrastructure and built environment (E.g. sea walls, resilient road networks, etc.)
☐	Ecosystem and ecosystem services (E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary (max 300 words)

In the past 50 years the Buenos Aires Province (PBA, for its acronym in Spanish) has suffered rainfalls of great intensity adversely impacting the Lujan River Basin where 11.4% of households fall within at least one of the unsatisfied basic needs indicators (a proportion larger than that of the Buenos Aires Province, which is 8.2%). During the last years the extent of flooding has increased resulting in waterlogging of soils in urban and rural areas as well as roads and trails. The general purpose of this Project is promote measures aimed at reducing and/or preventing river flooding, as well as controlling stream-flows and mitigating the impact of floods in the Lujan River Basin, thus causing more resilience *vis-à-vis* climate change in the basin.

Floods in the Lujan River Basin are mainly caused by: (i) meteorological and climate factors resulting in increases in rainfall frequency and intensity; (ii) physical characteristics of the basin such as: slope, contour, water-storage of the soil and water conveyance of the river; and (iii) anthropogenic factors such as changes in land-use, growing urbanization and the features of road and water infrastructure.

To address the issue, the Buenos Aires Province has structured the Lujan River Comprehensive Plan (LRCP) with an estimated cost of USD \$315,3 MM to solve the aforementioned problems. The plan envisages the development of an integral project to implement structural measures together with non-structural measures such as implementation of an early warning system (EWS), actions aiming at implementing land-use management and the strengthening of institutions. The scope of the project thus defined fully complies with the priorities set out by Argentina under the Intended National Determined Contributions (INDC) regarding climate change. Those priorities include implementation of early warning systems and application of structural and non-structural measures to face extreme events. Structural measures are the following: feasibility studies and civil works to increase the conveyance capacity of the river, construction of retention basins and substitution of inflatable dams for lock gates.

Funding of the project has been structured as follows: 29.13% PBA; 52.3% CAF and 18.56% GCF. This funding structure enables PBA to have the required cash flow and ensures that the project may be fully and successfully implemented within the timeframe agreed upon, ensuring the early implementation of structural and non-structural adaptation measures.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	- 19/07/2016 – Approval of first tranche from CAF - 04/07/2017 –Approval of second tranche from CAF
Expected financial close (if applicable)	30/9/2017 (considering the GCF approval)
Estimated implementation start and end date	Start: 01/07/2017 End: 30/06/2022
Project/programme lifespan	5 years 0 months

B.1. Description of Financial Elements of the Project / Programme

The Project and its financial sources:

- a. The Project's success is based on the integrated implementation of structural and non-structural measures as set out in the LRCP.
- b. The Project impact has been quantified as economic benefits brought about by the increasing resilience in the Basin, and, consequently, the avoidance of damages to dwellings in the Lujan River Basin resulting from less flooding.
- c. The Buenos Aires Province has resources available for partially funding the Project; however, for an optimal implementation, additional climate finance resources under concessional terms are needed. Further to ensure appropriate implementation of the Project, the financial structure should take into account the Province's future payment capacity.
- d. On 19 July 2016, CAF approved a first credit for an amount of USD \$100MM and a second credit was approved on July 4, 2017 for up to USD 120MM. The credit facility was divided into 2 stages due to limit of lending capacity from CAF to Argentina in year 2016 nevertheless it is fundamental that the Project is carried out as an integral Project.
- e. Against this backdrop, the financial structure proposed for the Project should be consistent with PBA payment capacity. To that end, the following is proposed:
 - i. Loan repayment and grace periods falling preferably in periods different from due dates for the payment of the current PBA debt.
 - ii. Concessional climate resources to support the PBA climate change mitigation efforts.
 - iii. Financing resources from the GCF that will support the PBA in its first climate change adaptation project.
- f. In addition to the loan, in order to replicate the Project, a grant of USD \$1.57 MM will be requested to the GCF.

Component costs of the Project:

Component	Sub-component (if applicable)	Amount (for entire project)	Currency	GCF Loan (USD Mill)	GCF Grant (USD Mill)	PBA (USD Mill)	CAF (USD Mill)
Component 1: <i>Engineering studies and other studies</i>	Sub-component 1.1: Engineering studies	1.18	<u>Million USD (\$)</u>			1.18	
Component 2: Water conveyance works - enlargement of the river channel and of water retention basins.	Sub-component 2.1: Widening Santa María channel	48.14	<u>Million USD (\$)</u>			17.08	31.06
	Sub-component 2.2: Enlargement of the Lujan River channel from RN # 6 to RN # 8	52.48	<u>Million USD (\$)</u>			18.62	33.86
	Sub-component 2.3: Enlargement of the Lujan River channel from RN # 8 until Belgrano bridge.	29.10	<u>Million USD (\$)</u>			10.33	18.77
	Sub-component 2.4: Enlargement of the Lujan River channel at Mercedes,	6.59	<u>Million USD (\$)</u>			1.50	5.09

		Sub-component 2.5: Enlargement of the Lujan River channel at Middle Basin,	52.94	<u>Million USD (\$)</u>			12.06	40.88	
		Sub-component 2.6: Enlargement of the Lujan River channel at Olivera	2.69	<u>Million USD (\$)</u>			0.61	2.08	
		Sub-component 2.7: Substitution of inflatable dams for lockgates	13.40	<u>Million USD (\$)</u>	10.35		3.05		
	Component 3: Construction of the retention basins and replacement of lock gates	Sub-component 3.1: ARTEH 1 to 7	47.96	<u>Million USD (\$)</u>	37.04		10.92	0.0	
	Component 4: Works for Bridges replacement and enlargement	Sub-component 4.1: Bridges to be demolished/ constructed	10.11	<u>Million USD (\$)</u>			3.11	7.00	
		Sub-component 4.2: Bridges enlargement	12.16	<u>Million USD (\$)</u>			4.16	8.00	
		Sub-component 4.3: Bridges clearance	0.39	<u>Million USD (\$)</u>			0.09	0.30	
	Component 5: Environmental and land-use issues	Sub-component 5.1: Environmental and land-use issues	13.28	<u>Million USD (\$)</u>			5.49	7.79	
		Sub-component 5.2: COMILU implementation	2.67	<u>Million USD (\$)</u>	2.06		0.61		
		Sub-component 5.3: Knowledge Dissemination	1.57	<u>Million USD (\$)</u>		1.57			
	Component 6: Early Warning System	Sub-component 6.1: Early Warning System	10.71	<u>Million USD (\$)</u>	7.52		0.71	2.48	
	Component 7: Strengthening,	Sub-component 7.1: Strengthening,	8.53	<u>Million USD (\$)</u>			2.34	6.19	

	supervision and auditing	supervision and auditing							
	Component 8: <i>Administrative Cost for CAF Financing</i>	Sub-component 8.1: <i>Administrative costs for CAF Financing</i>	1.46	<u>Million USD (\$)</u>				1.46	
	Total project financing		315.36	<u>Million USD (\$)</u>	56.97	1.57	91.86	164.96	

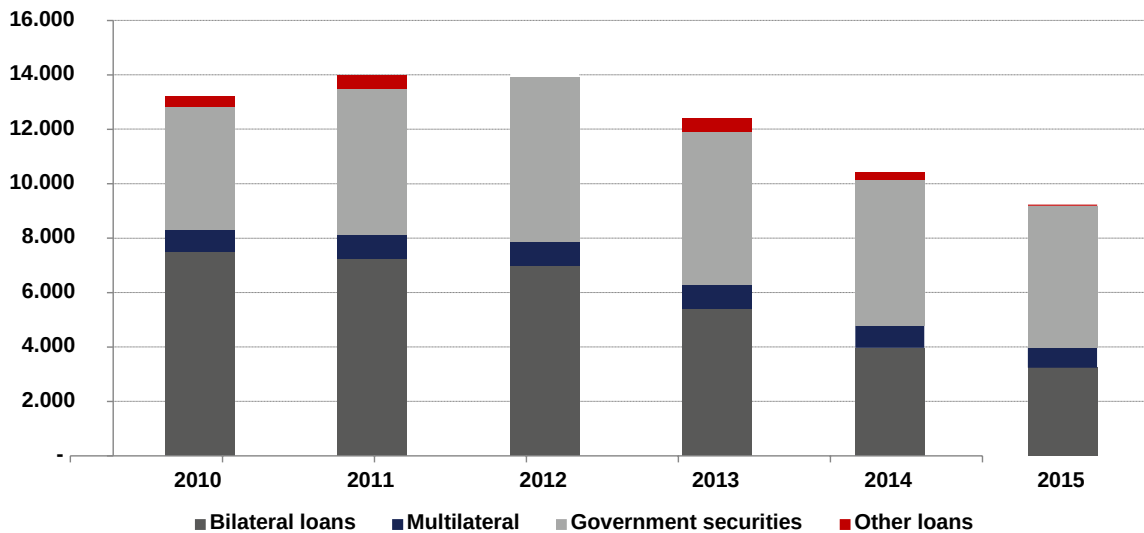
* Please expand the table if needed.

B.2. Project Financing Information

	Financial Instrument		Amount	Currency	Tenor	Pricing	
(a) Total project financing	(a) = (b) + (c)		315.3				
(b) GCF financing to recipient	(i) Senior Loans		56.96	<u>Options</u>	(20) years	(0.75) %	
	(ii) Subordinated Loans				5 years of grace	() %	
	(iii) Equity				() years	() % IRR	
	(iv) Guarantees						
(b) GCF financing to recipient	(v) Reimbursable grants *		1.57	<u>Options</u>			
	(vi) Grants *						
<i>* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of concessionality should correspond to the level of the project/programme's expected performance against the investment criteria indicated in section E.</i>							
Total requested (i+ii+iii+iv+v+vi)				<u>Options</u>			
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority
		100.0		CAF	(15) years		
		64.9**		CAF**	54 months grace period	(Libor +2) %	<u>Options</u>
					(15) years	(Libor +2) %	<u>Options</u>
		91.9		<u>Options</u>	<u>PBA*</u>	54 months grace period	() % IRR
Lead financing institution: CAF							

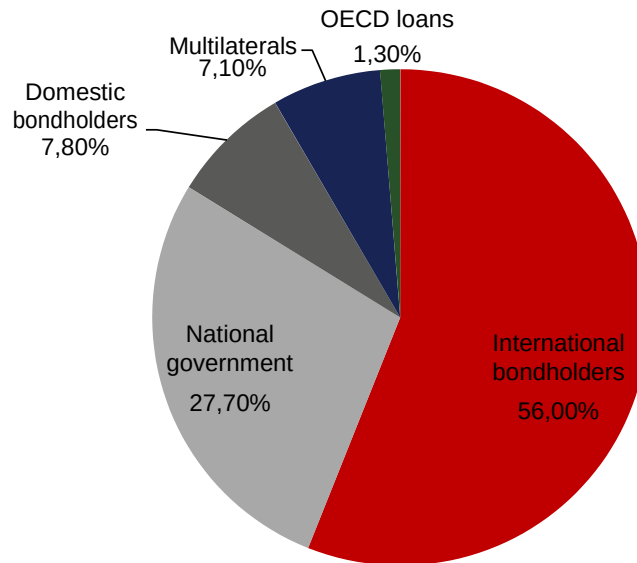
	*PBA will finance USD (\$) 91.9 Million from its budget, it was included as Equity in Co- financing to recipient ** AFD is assessing its participation in the co-financing of the Project for up to USD \$30 million. As such, up to USD \$30 million of the second tranche of CAF's senior loan may be assigned to AFD * Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.				
(d) Financial terms between GCF and AE (if applicable)	In cases where the accredited entity (AE) deploys the GCF financing directly to the recipient, (i.e. the GCF financing passes directly from the GCF to the recipient through the AE) or if the AE is the recipient itself, in the proposed financial instrument and terms as described in part (b), this subsection can be skipped. If there is a financial arrangement between the GCF and the AE, which entails a financial instrument and/or financial terms separate from the ones described in part (b), please fill out the table below to specify the proposed instrument and terms between the GCF and the AE.				
	Financial instrument	Amount	Currency	Tenor	Pricing
	Choose an item.		<u>Options</u>	() years	() %
	Please provide a justification for the difference in the financial instrument and/or terms between what is provided by the AE to the recipient and what is requested from the GCF to the AE.				
B.3. Financial Markets Overview (if applicable)					
PBA financing sources As of 31 March 2016 the Buenos Aires Province's public debt of amounted to USD \$10.431,8 MM. However, PBA dollar-denominated debt has been decreasing since 2012. This deleverage is the result of debt reduction with the national government: in 2010, the debt with the national government made up 56% of total debt, whereas, by 2015 this ratio was 35%. The evolution of the debt in the past ten years shows that its proportion with respect to the Provincial economy is decreasing. The Debt/GDP ratio has continuously decreased from 13.6% in 2006 to 6.3% in 2016. Notwithstanding, it is expected that this trend (existing since 2002) is going to change starting in 2016, enabling PBA to finance new projects in the future.					

**Evolution of the PBA's total indebtedness per type of debt
(US\$ MM)**



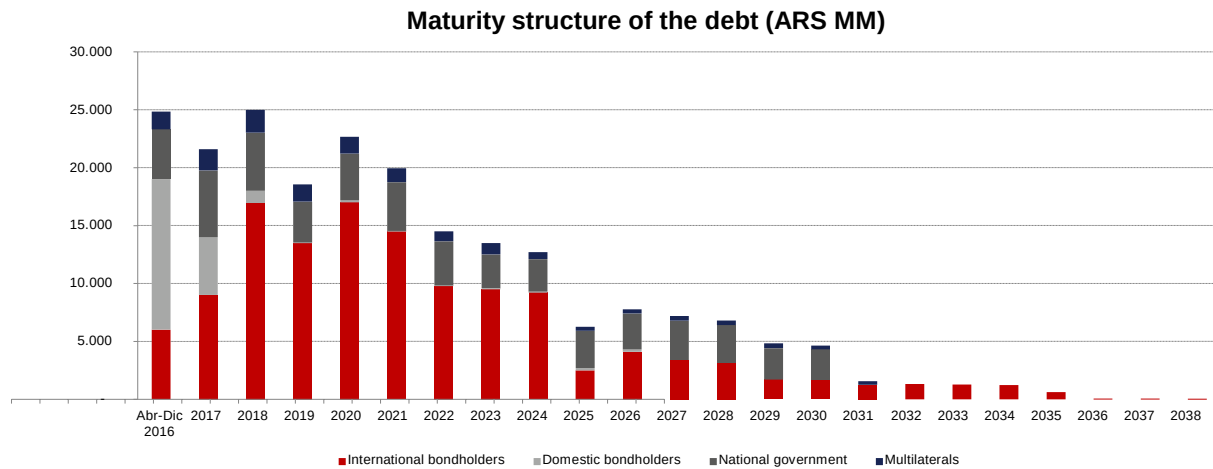
BPA financing sources as of 31 March 2016 are the following: international bondholders (56%), national government (28%) and domestic bondholders (7,8%).

Debt composition



Source: Provincial Directorate of Indebtedness and Public Credit – Finance Subsecretariat

The maturity structure of the aforementioned debt is the following: mid term 2017-2021 (50%), long term 2022-2028 (39%) and short term 2016 (11%).



On 16 March 2016, PBA issued debt in the international capital markets for an amount of USD \$1.250 MM to fund infrastructure projects. The Province issued 8-year maturity bonds (due date in 2024) and an annual coupon of 9,125%.

International bondholders have been the main source of funding for the Province. This source of funding has two constraints: i) the repayment period: near 60% of the outstanding debt's due date is earlier than 2021; ii) the cost of financing: the last debt issuance had a cost of 9,125%.

Access to climate financing resources such as GCF will offer the possibility to PBA to implement climate change adaptation measures which will result in positive social, economic and environmental benefits for the region

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

C.1. Strategic Context

The Republic of Argentina is made up of 23 Provinces and the autonomous city of Buenos Aires. Argentina has a population of nearly 40 million inhabitants, 90% of which is located in urban areas. The Metropolitan Area of Buenos Aires, which includes the city of Buenos Aires and 26 municipal jurisdictions, accounts for 0.5% of the national territory and concentrates 31.92% of the population.

Between 2004 and 2008, Argentina was a leader in poverty reduction, but since 2008 this trend has slowed down. In 2014, 12.7% of the population was considered poor (living with less than USD \$4 a day). One third of the population of Argentina lives with an income between USD \$4 and USD \$10 a day, and are at risk of falling into poverty.

In the last decade the main concern of the Federal Government was economic recovery and economic growth. However, this policy brought about some important distortions in the economy. The new administration, inaugurated in December 2015, is introducing adjustments to eliminate economic distortions responsible for hindering economic growth and for increasing the inflation rate. Argentina is now living in a context of changes expected to bring about results in the mid term with the reduction of fiscal deficit and increases of debt capacity later on.

The Republic of Argentina and the Climate Change:

In 1993 Argentina ratified the United Nations Framework Convention on Climate Change (UNFCCC) and, in 2001 the Kyoto Protocol. In 2002, a presidential decree replaced the Secretariat of Environment and Natural Resources with the Secretariat of Environment and Sustainable Development under authority of the Head of the Cabinet of Ministers. By December 2015 it became the Ministry of Environment and Sustainable Development vested with significant institutional duties. Its core purpose is to provide assistance to the President and the relevant Ministries, as well as «*coordinating all and any public policies having impact on the environmental sector, by implementing a strategic planning of environmental policies and programmes adopted by the national government*». The new Minister relies on the Secretary for Environmental Policy, Climate Change and Sustainable Development that hosts the Directorate for Climate Change and is responsible for the adoption and coordination of policies and actions relating to climate change. Additionally, there is a government committee for climate change with the involvement of both Federal and Provincial governments.

In the Third National Communication to the United Nations Framework Convention on Climate Change (November 2015), Argentina highlighted the country's vulnerability *vis-à-vis* climate change, analyzed greenhouse gas emissions (GHG) and put forward a set of planned mitigation measures and the need for adaptation measures to address problems created by frequent heavy rains, such as floods, with subsequent harm to the population and the agricultural production.

Furthermore, prior to the Paris negotiations in 2015, Argentina submitted to the UNFCCC, its Intended Nationally Determined Contributions (INDC), where it made clear that adaptation to climate change is a major concern, whereby it has implemented adaptation measures. Nevertheless, to further such adaptation actions, international financing, technology transfer and institutional strengthening are required. Two out of the main seven adaptation measures considered in the INDC include early warning systems and implementation of structural and non-structural measures to face extreme events.

Additionally, the Inter-jurisdictional Cabinet on Climate Change was established on July 25 2016 for the purposes of articulating policies on climate change and increasing the society's awareness of their relevance and significance. Among these core duties is aligning the national and provincial governments with regard to actions on mitigation and adaptation to climate change; propose a National Plan on Response to Climate Change, as an input to discuss a core comprehensive national strategy and contribute to capacity-building in the field of prevention and handling of emergency and disaster situations arising from extreme climate events, among others.

Regulatory Framework in Argentina

For the protection of the environment Argentina has in place a regulatory framework both at the federal level and the provincial level.

National Framework:

In 1994 Argentina introduced in its Constitution the obligation to protect the environment granting to its population the fundamental right of a healthy environment. In addition, the Constitution gave the Provinces the right to regulate and introduced the notion of “Minimal Budgets for Protection” to ensure a minimal threshold of environmental quality across the national territory. The two laws currently enacted on “Minimal Budgets” for the Lujan River Project are the following:

- Law 25.675, General Law on Environment (Law 25.675). This Law sets out the notion of land-use as a tool for environmental management. It also establishes guidelines for the environmental policy and the Congruence Principle to align the domestic legal framework with the goals and targets of sustainable development.
- Law on Minimal Budgets for the Integrated Water Resources Management (Law 25.688). This Law is the basis for creating agencies devoted to river basin management.

At the national level, the Federal Agreement on Water Resources and the Guiding Principles for the Integrated Water Resources Management (Law 26.438) seek integrated management of water resources through the notion of territorial areas in river basins. The Province of Buenos Aires has already ratified this Law.

Provincial Framework

At the Provincial level there is a large number of regulations for the integrated river basin management:

- Law 11.723 (amended by Law 13.516) is the General Law on Environment, which is the environmental regulatory framework for the Province of Buenos Aires. This Law sets out guiding principles governing the Province’s environmental policy. The purpose of this Law is “the protection, preservation, improvement and restoration of natural resources and environment in the Province of Buenos Aires, to preserve life in the broadest sense, ensuring present and future generations the preservation of biological diversity and the quality of the environment”.
- Law 5.965, which sets out the framework for water resources protection. This Law is the basis to fight against water pollution in the Province and to grant wastewater discharge permits into receiving waters.
- Law 12.257 (Water Code). The Water Code of the Province of Buenos Aires, with its amendments, is the most important piece of legislation for water regulation and water management for river basin units in the Provincial territory.
- Law 14.710, which creates the Committee of the Lujan River Basin Committee (COMILU). This is a major body for the implementation of the Lujan River Project.

Additionally, with regard to planning and management of water resources, the government of the Province of Buenos Aires approved in 2007 the Provincial Water Management Plan contemplating differentiated actions for the metropolitan area, the rural area and the coastal area. The plan calls for investments of USD \$1.000 millions which benefit 8 million of people in Lujan, Salado, Reconquista, Matanza-Riachuelo, Areco and Rio V river basins and the region adjacent to the capital (La Plata, Berisso and Ensenada).

Institutional Context - CAF:

CAF’s involvement in the Project would be aligned with four of its institutional objectives:

- i. CAF’s Adaptation to Climate Change Program which promotes and supports planned processes of adaptation to climate change in Latin American countries, at different levels such as policies, plans, programmes, and projects to promote their sustainable development
- ii. CAF’s green financing, which accounted for 24% of approvals in 2015, is expected to increase to 30% by 2020.
- iii. CAF’s financial support coupled with USD \$58.5 MM from GCF facilitates the implementation of the full program before expected, promoting early adaptation measures in the Lujan River Basin and securing completeness in the implementation.

CAF's experience in Argentina starts in 2010. In these years 2010-2014, CAF's operations were aimed at providing ongoing technical and financial support to several sectors: electricity, logistics, water and sanitation, and education for a total of USD \$5.566 MM approved.

C.2. Project / Programme Objective against Baseline

In the last century the Southeast region of South America experienced changes in the rainfall pattern affecting the water balance of the region, particularly in La Plata River Basin.

Floods in the Lujan River Basin have both direct and indirect impact over near 8 million people of the Province of Buenos Aires. In a timeframe of two years around 10.017 precarious dwelling became recurrently flooded and it is estimated that in a timeframe of 100 years this figure would rise to 15.085. The floods affect mainly single-family dwelling (87% of floods).

Recurrent Floods						
Property	2 years	5 years	10 years	25 years	50 years	100 years
Single family dwellings	8.722	9.983	10.625	11.629	12.373	13.189
Multi-family dwellings	353	468	491	589	627	666
Precarious dwellings	163	173	187	195	211	221
Industry	111	121	128	135	140	145
Commerce	668	726	762	808	837	864
Total	10.017	11.471	12.193	13.356	14.188	15.085

The Lujan River Basin is large and narrow. Water from heavy rains reaches the main river course very quickly, thus raising abruptly the river's water level and bursting the banks of the river. Floods in the river basin occur rather fast, on a large-scale and unexpectedly while people become defenceless in face of such natural phenomena.

Over time this problem has become more acute due to the following reasons:

- Climatic Factors.
- Physical Factors.
- Anthropogenic Factors.

Climatic Factors

Rainfall and temperature are the main variables influencing the region's climatic characteristics. In wet years, the increase in rainfall results in a disproportionate increase in the streamflow because water infiltration (absorption capacity of soil and subsoil) is low and water evaporation decreases. On the other hand, in dry years, the reduction in rainfall results in a proportionally larger reduction of streamflow.

In short, rainfall variations in the zone result in a magnified river basin response. The lack of water brings about restrictions in water provision for both human consumption and agriculture, whereas the excess of water results in floods.

Streamflow is also affected by the spatial distribution of rainfall. The larger the rainfall intensity and frequency over a zone of the river basin with low interception capacity and/or over a zone with good drainage capacity, the larger the magnitude of river flooding.

The Research Centre for the Sea and the Atmosphere (CIMA) made a climate forecast for Argentina using two timeframes: near future (2015-2039) and distant future (2075-2079). The study concluded that – under different

recurrences – the daily maximum rainfall will increase 12 mm. and the maximum accumulated for 5 days will increase 18 mm

In addition to the previous results, the Lujan River Basin study analysed the increase in extreme precipitations for the Lujan basin. This study took in account the historic maximum precipitations for 1 day and 5 days accumulated from the period between 1961 and 1990- It also projected medium term (from 2015 until 2039) and long term (from 2075 to 2099)

In the Higher basin, for the medium term, the results for 1 day precipitation show increases of maximum precipitation between 10% (CMCC-CM and NorESM1-M models) and 27% increase (CCSM4 model). Model MRI-CGCM3 shows a potential decrease of 6% for the 1 day maximum precipitation. For the 5 days accumulated trends are moderate. With regards to the long term, all studies show increases between 13% and 38% for daily precipitation and 24% to 29% for 5 days accumulated precipitations.

In the lower basin, all models predict for the medium and long term similar moderated increases.

Physical Factors

During 1989-1994 conventional farming in the river basin was based on terrace farming. Thereafter, the region changed the modality of land preparation for direct seeding (cereals, oilseeds). This type of land preparation for cultivation modifies the runoff of the surface whereby a larger part of the rainfall infiltrates, resulting in a larger subsurface runoff. Intensive use of land for agricultural purposes gives rise to soils with sparse vegetation and low level of interception resulting in an acceleration of the runoff and a reduction of temporary retentions. As a result, the volume of water in the river becomes larger in shorter spans of time.

On the other hand, channelization in the river basin should rely on a technical justification and the subsequent approval from provincial or municipal authorities. Clandestine channelization generally lacks of technical planning, and adversely affects the drainage network of creeks, channels and rivers.

Anthropogenic Factors

Urban growth brings about a decrease in interception capacity and an increase of soil impermeabilisation, which increases the volume of water and the speed at which such water flows into river courses. In addition, the increases in residential density calls for more infrastructure works, more roads, more public utilities, etc., in other words, changes in land-use increase the levels of interception of runoff and, also, increase vulnerability in the region.

Objective of the Project

The objective of the Project is to prevent flooding, control stream-flows and mitigate the adverse effects from floods in fifteen municipalities of the Lujan River Basin. In order to achieve this objectiv it is required to solve problems in the river basin derived from physical and anthropogenic factors and, implement other non-structural measures.

The following are the core measures to be implemented to solve the causes of flooding and, thus, increase resilience capacity:

Cause of the Problem	Measures to be Implemented	Impact
Physical Factors		
Low water conveyance of the river	Implementation of supplemental channels and enlargement of natural river courses Works included in Component 2 that will be complemented by works included in Component 3 (ARTEH). Previous to its implementation studies in Component 1 need to be carried out.	Mitigation or reduction of the discharge derived from the storage in the retention area. Delay flooding timing through displacement or temporary delay of maximum peaks.

Current conditions of the river basin runoff	Replacement and enlargement of bridges, crossings and lockgates. Works included in Component 4. In order to improve runoff conditions, works in Component 2 and 3 are needed. Previous to its implementation studies in Component 1 need to be carried out.	Improvement of current runoff conditions by removing river obstructions and detritus in middle and lower parts of the basin.
Anthropogenic Factors		
Urban growth in cities resulting in an increase of population density.	Implementation of a territorial management framework suitable for water, environmental and socioeconomic needs of the river basin. Included in Component 5. In order to maintain the impact of Components 2, 3 and 4, adequate land use regulation needs to be enforced	Improvement of housing conditions and life standards of families to be relocated.
Other Concerns		
Unexpected large-scale floods	EWS Works included in Component 6	Improved prediction tools and anticipation to extreme events, thus increasing adaptation capacity
Unintegrated river basin management	COMILU Included in Component 5- Key to the Correct implementation of the complete Project.	Appropriate management of all aspects of the river basin. Measures implemented in the Project should be sustainable in the future.

The Project will reduce floods, thus improving economic conditions in the region, i.e. improved conditions for agricultural production, industrial production and commerce, as well as improved conditions for people in the region. Additionally, implementation of the Project will mitigate damages in dwelling located in land with recurrent floods. Indeed, the proportion of damages avoided by the Project with respect to damages without the Project is near 58%, as shown in the table below.

Recurrence (years)	Without project (USD \$)	With project (USD \$)	Damages avoided (USD \$)	
2	87.937.831	35.577.984	52.359.847	60%
5	106.208.246	43.278.805	62.929.440	59%
10	116.520.213	49.583.578	66.936.635	57%
25	130.708.645	58.186.311	72.522.334	55%
50	145.953.798	63.111.719	82.842.078	57%
100	165.687.864	71.428.936	94.258.929	57%

Near 1.6 million people from the municipalities of Campana, Escobar, Exaltación de la Cruz, José C. Paz, General Rodríguez, Malvinas Argentinas, Moreno, Pilar, San Fernando and Tigre, living in the area of influence of the Project in the middle and lower parts of the basin, will draw direct benefits from the Project. It is estimated that around 1.2 million people, that is, the remaining inhabitants of the above-mentioned municipalities and people living in the upper part of the basin, will indirectly benefit from the Project. This said, a total of 2.8 million people representing 17% of the total population of Buenos Aires Province would benefit from the Project.

C.3. Project / Programme Description

Describe the main activities and the planned measures of the project/programme according to each of its components.

Conceptualization of the Project

The Ministry of Infrastructure of the Buenos Aires Province through the Provincial Directorate of Water Infrastructure (DPOH) conducted the “Study of the Integrated Plan and Project for the Construction of Infrastructure for the Regulation and Sanitation of Lujan River”. This study provided the planning of structural and non-structural measures that, together with the basin systematization, will serve to mitigate flood impact.

The study analyzed 8 structural measures from environmental, social, economical and financial point of views, by calculating such measures cost-effectiveness and choosing the necessary works for protecting and ensuring the quality of life of the population and, thus, consolidating urban and commercial development of the region.

The non-structural measures consist of building an institutional framework to: ensure the coordination and sustainability of floods risk management; benefit from technological developments in climate forecasting in order to improve prevention actions; involve the population into awareness campaigns through civil society and to obtain prompt responses in case of disasters through groups of first responders.

Likewise, the objective of the Project is consistent with the purpose of Law 14.710 “COMILU” Committee of the Lujan River Basin enacted on 5 May 2015. The COMILU is an autonomous entity with legal capacity to act both under private and public law and which objective is to improve the quality of life and the environmental and health conditions in the Lujan River Basin.

Components of the Project

The Project has 8 components. Components 1, 2, 3 and 4 seek to address problems derived from physical factors; component 5 seeks to address problems derived from anthropogenic factors adversely affecting the river basin and component 6 seeks to mitigate the issues relating to abrupt occurrence of floods.

Component 1: Engineering studies and others: Performance of executive projects and preparation of tender documents for civil works under the Project needed in order to finalize engineering designs and to prepare the tendering process for the engineering works considered in Components 2, 3 and 4.

Component 2: Water conveyance works and enlargement of the river channel. These will improve the conveyance capacity of the Lujan River through construction of supplemental channels, enlargement of channels and crossing works:

- Santa María channel, communicating Lujan river with Parana de las Palmas River, will increase its cross-section over a distance of 7.1 Km.
- Enlargement of Lujan River channel with trapezoidal sections over a distance of 12 Km, from provincial route (RP) 6 until the bridge of the national route (RN) 8 in Pilar.
- Enlargement of Lujan River channel with trapezoidal sections over a distance of 9.5 Km, from RN 8 until the railway bridge of Belgrano North line.
- Improvement of the river conveyance capacity through enlargement of its cross-section in the locality of Mercedes, in the zone of Olivera, and in a section corresponding to the locality of Lujan (15.5 Km).

Component 3: Construction of retention basins and replacement of lock gates. Construction of areas for temporary retention of surplus water (ARTEH)- This component is complementary to Components 2 and 4. The retention basins, built on the higher part of the river, are meant to remain empty, when the river floods they retain water thereby allowing lower parts of the river to evacuate its excess water.

- Areas for temporary retention of surplus water (ARTEH) in the following creeks: Los Leones, Mayan, Leguizamón, Grande, del Oro and other two over the Lujan River.
- Substitution of inflatable dams for lock gates in Lujan and Mercedes.

The lockgates replacement will allow adequate waterconveyance thereby being complementary works of Component 2.

Component 4: Works for bridges replacement and enlargement. Works for replacement and enlargement of bridges to improve current runoff conditions, removing river obstructions and detritus. These works will be in addition to the construction of channels. This component is complementary to Component 2 and needs to be implemented in order to fully obtain the benefits of enlarging the channels.

The work will be performed over 15 bridges across the river basin. Bridges will be enlarged or replaced depending on the need of enlargement and their structural conditions.

Component 5: Environmental and land-use issues. Establishment of the riverbank to define boundaries on private and public areas and identification of flooding zones that will be subject to administrative restrictions or easements. The above entails review and adaptation of local land-use plans, identification of protected areas and, consequently, there will be grounds for expropriation and new settlements. The land acquisition framework outlines the policies and procedures which will apply to this process. This gives rise to tools for delimitation of floodplains as an input for strategic territorial development plans in the concerned municipalities. This component is key to the integrated management of the Lujan basin and is one of the main non- structural components that complement the structural ones explained above.

Component 6: Early Warning System. The EWS will be integrated into the Provincial Early Warning System of the Ministry of Infrastructure and Public Services of the Province, through the Directorate for Provincial Hydro-environmental Monitoring, an organizational unit of the Sub-secretariat of Water Infrastructure. The EWS of the Project has four systems:

- **Monitoring and Surveillance System:** consolidates information relating to weather and rain forecasts; responsible for the measurement and transmission network; responsible for data processing and follow-up to extreme events.
- **Warning and Alarm System:** fixes alert threshold levels and develops actions in case of triggering any alert or warning, thus alerting the authorities and the community.
- **Communication System:** responsible for communications with the National Meteorological Service, the measurement and transmission network, and institutions and stakeholders linked to the alert and evacuation system.
- **Evacuation System:** EWS operators' efficacy to transmit alerts of extreme events to those persons responsible; skills of people involved in implementing the Evacuation Plan; community knowledge over steps to follow in case of evacuation.

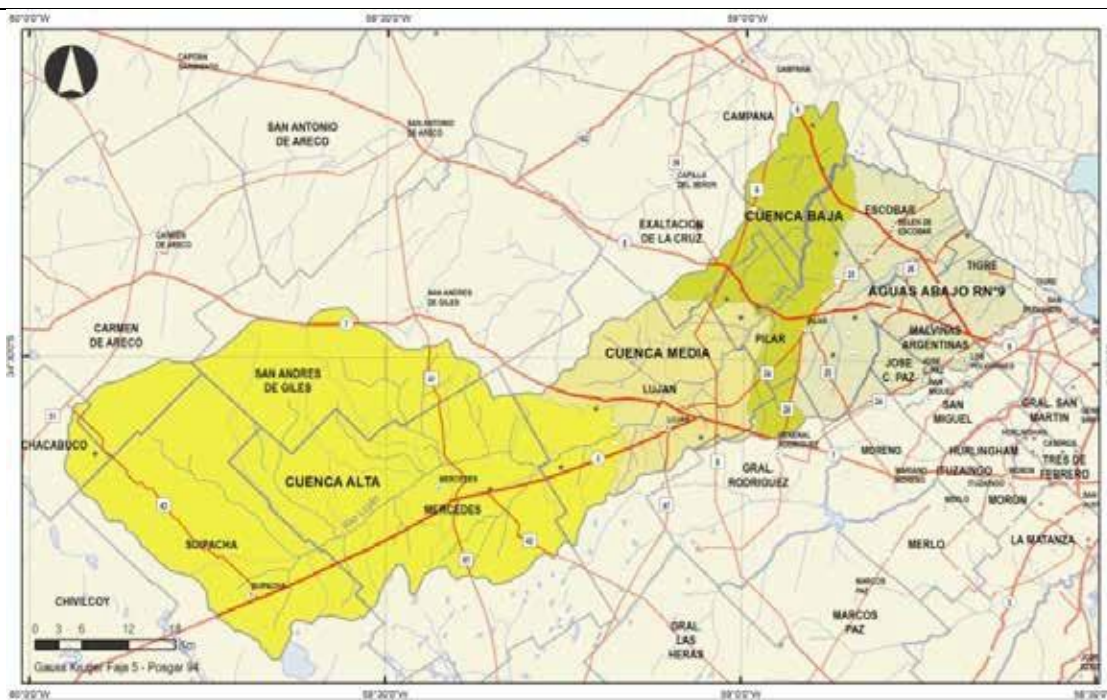
This is one of the non- structural measures necessary to develop an integrated project that responds to adaptation needs.

Component 7: Strengthening, supervision and auditing. This component includes strengthening all institutions dealing with the Project implementation, mainly COMILU, through consulting services, installation of furniture and IT equipment, and operational expenditures. It also includes hiring technical, social, and environmental supervision of works, as well as the external audit for the Project. The supervision included in this component relates to external supervision that will be hired by the PBA in order to ensure the correct implementation of the project.

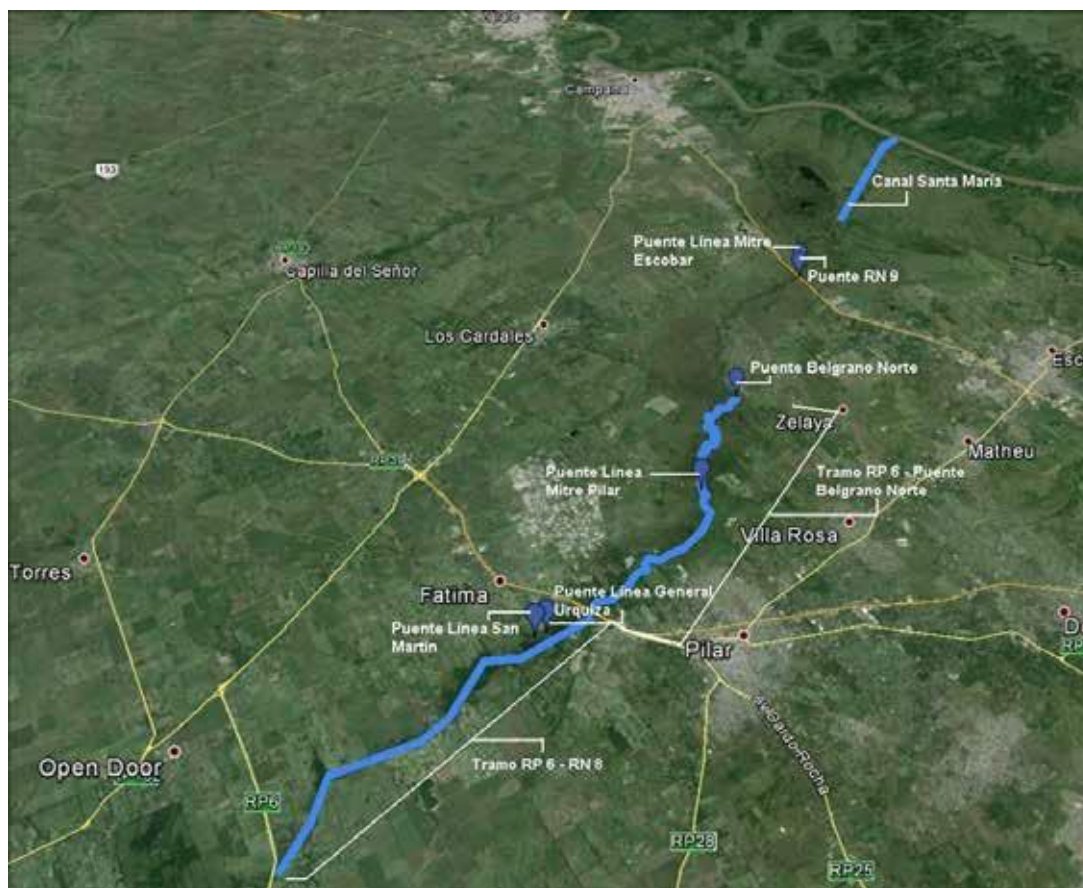
This component, also non- structural will, through the COMILU, support policy implementation and the correct land use along the Lujan River that will allow the defined measures to deliver as planned.

Component 8: Administrative Cost for CAF Financing. Include evaluation expenses and funding fees over the CAF's funding (only applicable for CAF's resources, this not include CAF's fees as GCF accredited entity).

Maps:



Map: Lujan River Basin



Map: localization

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

Province of Buenos Aires (PBA) Sponsor of the Project

PBA has a wide experience in implementing projects and programmes funded through multilateral organizations. In the past ten years, the government of the Province has implemented projects and programmes of international financing for an amount of over USD \$1.000 MM in sectors such as drinking water supply, sewage and draining networks. PBA has ongoing projects for an amount of USD \$570 MM and over USD \$1.500 MM in the pipeline. In the last 5 years PBA has implemented an average of USD \$92 MM per year in water and sanitation projects.

PBA has in place an appropriate regulatory and institutional framework, with well-defined functions and roles for all stakeholders involved in the planning, implementation, follow-up and operation of the Project. Additionally, the Project contemplates institutional strengthening of COMILU, responsible for the coordination of the LRCP implementation and for the integrated management of the river basin. For this reason it is important that non-structural measures provided for in the Project accompany the implementation of structural measures to effectively reduce vulnerability of the population.

PBA has skilled technical and administrative personnel in Ministries, Sub-secretariats and Directorates in connection with the management of the operation. The Project can also count on the cooperation of institutions at national, provincial and municipal level.

C.5. Market Overview (if applicable)

Not applicable.

C.6. Regulation, Taxation and Insurance (if applicable)

At present, it is impossible to provide a list of all licences and permits required, as the detailed definition of the Project is part of Component 1. In any case, for its implementation, the Project must comply with all regulations from the Republic of Argentina and the Providence of Buenos Aires.

Notwithstanding the foregoing, prior to the first disbursement, CAF has requested the submission of a state of progress of formalities for the obtention of licences and permits, thus ensuring compliance with all requirements. Specifically, CAF's resolution state that: (i) prior the first disbursement, the project will submit an initial report including, among other aspects, status of processing of environmental licenses and any other permits required, and action plan (next steps) to securing them; (ii) prior to the start of each work the Project will submit a copy of environmental permits and/or licenses as granted for Works to get under way and / or each Works stage.

With regard to taxes, the Project must pay Federal and Provincial taxes for projects with the characteristics herein contained.

C.7. Institutional / Implementation Arrangements

Ministry of Economy of the Buenos Aires Province through the Subsecretariat of Finances (SSF, for its acronym in Spanish), will be the Borrower's Representative, with financial management functions for the Loan's resources. The SSF's basic responsibilities will be:

- i. Act as a direct counterparty to the Bank within the boundaries of its purview.
- ii. Underwrite the administrative acts that are necessary pursuant to the Program's execution and provide for the timely fulfillment of the Prior Conditions to the first disbursement and other sundry conditions in the contract which correspond to the SSF.
- iii. Maintain the registry and record of the contracts and their modifications, with their corresponding administrative actions, that are financed wholly or in part with resources derived from the loan.
- iv. Record the tangible progress of the contracts, through certificates of works and others, and corresponding backing documentation for goods and services.
- v. Administrate the financial resources of the loan, process the disbursements and maintain accounting records of the Program's financial execution.
- vi. Contract the Program's Audit.

Implementation: Given PBA's experience of administration in the management of external financial resources, the Project will be implemented through the current structure of the provincial government.

Ministry of Infrastructure and Public Services of the PBA (MISP), as competent body for the sector, it will be technically responsible for the implementation and follow-up of the Project. The entity responsible for the project's execution is the MIPS which will act through the Works Projects Coordination and Execution Unit (UCEPO, for its acronym in Spanish). In turn, the UCEPO will rely on technical assistance from the following bureaus of the MlySP:

- Subsecretariat of Hydraulic Infrastructure, through three (3) offices relevant to the project's progress:
 1. Provincial Office of Public Works (DPOP, for its acronym in Spanish).
 2. Provincial Office of Maintenance (DPM, for its acronym in Spanish).
 3. Provincial Office of Hydro-environmental Monitoring (DPMHA, for its acronym in Spanish)
- Administrative Subsecretariat through the Provincial Office for Acquisitions and Contracting (DPCyC, for its acronym in Spanish).

Work Projects Coordination and Execution Unit – UCEPO. The Program's General Coordination corresponds to the UCEPO, and it is responsible for the opportune financing of the local contribution and for the fulfillment of the Program's objectives and the execution of the components, in the timeframes and with the budget provided in the Loan Agreement.

The basic responsibilities as executor will be:

- i. Act as a direct counterparty before the Bank within the boundaries of its purview.
- ii. Coordinate the execution of the Program, with the distinct areas of the MlySP in order to achieve compliance with the objectives of its components, in the timeframes and budget provided for in the Loan Agreement.
- iii. Supervise and followup the development of each one of the components, verifying the due execution of the cycle of each project.
- iv. Undertake the actions necessary to fulfill the clauses of the Loan Agreement.
- v. Elaborate the basic documents for scheduling, monitoring and evaluation of the Program (Annual Operating Plan, Acquisitions Plan, Quarterly Reports);
- vi. Approve the Program's Operating Manual and its modifications, prior to an agreement with the SSF.
- vii. Provide for the timely fulfillment of the Prior Conditions to the first disbursement and other sundry conditions of the contract, with corresponds to the OE.
- viii. Generate the Bid Documents, acquire the goods and contract the consulting and non-consulting services of the Program.
- ix. Supervise the application of the procedures related to the contracting of works for the Program including the content applicable to its socio-environmental management.
- x. Manage the corresponding payments in the framework of the Program's components.
- xi. Carry out the periodic followup on the compliance with the products and timeframes of the projects and components in accord with the results matrix, schedules and other followup instruments for the Program.
- xii. Manage the performance of project evaluations.
- xiii. Request from the SSF the availability of Loan funds necessary for carrying out the Program's payments.
- xiv. Provide assistance to the intervening technical areas in order to ensure compliance with the environmental and social safeguards required by the Bank.

Subsecretariat of Hydraulic Infrastructure – SSIH. With regard to technical support, as was previously indicated, the UCEPO will have the technical support of other relevant areas which, just as the UCEPO, form part of the MlySP. Said support comes from the Subsecretariat of Hydraulic Infrastructure through three provincial bureaus: Public Works, Hydro-environmental Maintenance and Monitoring. The roles of each one of these bureaus are identified below:

- Provincial Office of Public Works will have the following responsibilities:
 - a) Prepare the TPS and ToR of the actions and activities required for the execution of the Program's actions.

- b) Prepare and approve work progress certificates.
- c) Act, when necessary, as the direct counterparty for the consulting services and for goods, ensuring the compliance of the Program's objectives.
- d) Carry out the supervision of inspection services of the works.
- e) Request from the UCEPO the availability of the necessary Loan funds, to order the payments for the corresponding works.
- f) Receive both in a provisional and definitive manner the Program's works.
- g) Underwrite, when necessary, the agreements of works transfers.
- h) Maintain the UCEPO duly informed about the execution of activities and likely risks and problems.

- Provincial Office of Maintenance.

Its activities will report to the Subsecretariat of Hydraulic Infrastructure, in coordination with the DPOP [Provincial Office of Public Works].

Responsible for implementing and executing the scheduling of the execution of the hydraulic works that the SSIH [Subsecretariat of Hydraulic Infrastructure] as regards the maintenance, improvement and expansion of all the hydraulic infrastructure.

Intervene and resolve the agreements covering the provision of equipment and the execution of tasks within the scope of the SSIH with Municipalities, Basin Committees and Consortiums.

- Provincial Office of Hydro-environmental Monitoring

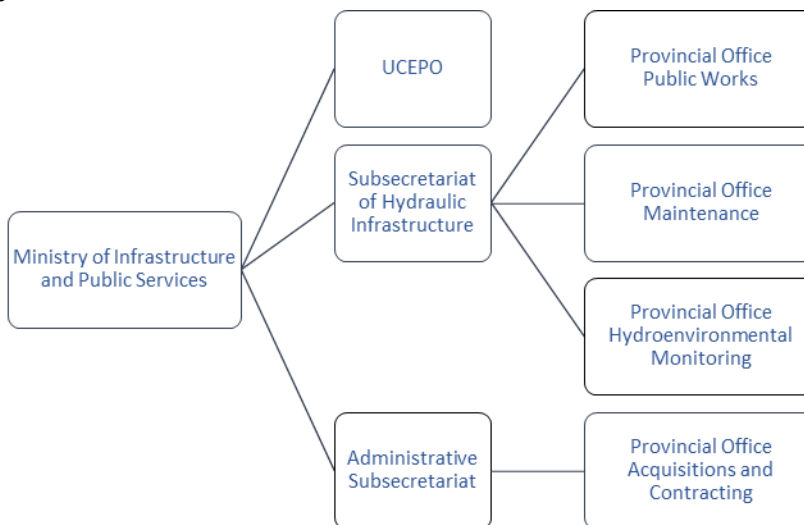
Is in charge of promoting the management of water resources and the identification of hydro-meteorological risks. In addition, supervising the production and processing of the hydrological data and issue reports, statistics and all other information.

In turn, coordinate the provision of the alerts and risks in a timely manner to the distinct entities.

Undertake works with the ADA [Water Authority, acronym in Spanish] in order to prevent and mitigate water contingencies and develop technical criteria and protocols for the design and operation of early warning systems in the face of floods.

Administrative Subsecretariat. The Administrative Subsecretariat will be supporting the development and execution of the Project through the Provincial Office of Acquisitions and Contracting from where the contracting processes for the Project works will be done, at the request of the intervening offices. This Provincial Office has two areas: (i) office of goods and services, and, (ii) office of contracting of public works.

In summary, from the MIySP:

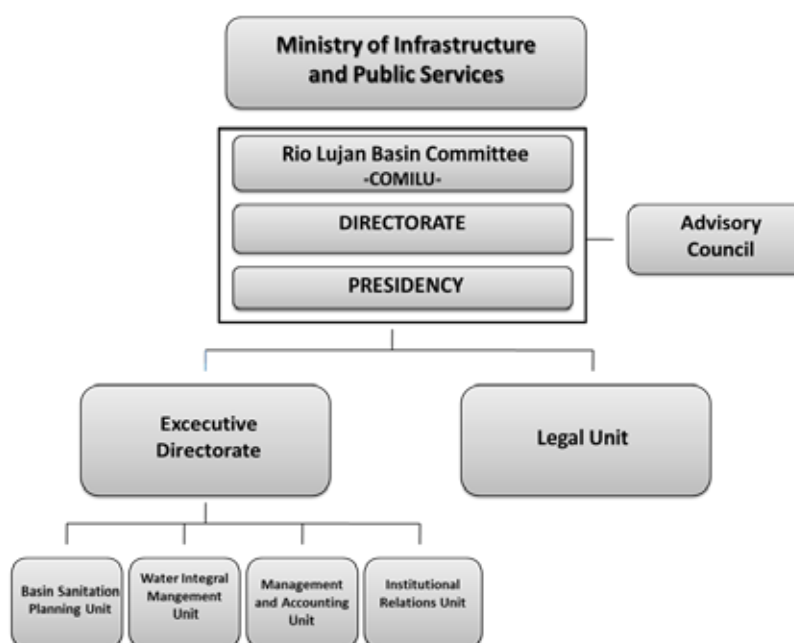


Technical and Environmental Inspection of Civil Works: an external consulting firm should be hired; this firm will be supervised by the SSIH team of inspectors.

Financial Auditing: an external auditing firm should be hired.

COMILU: It will be established as River Basin Committee to strengthen the integrated management of the basin. The core functions of the River Basin Committee will be the formulation of public policies and strengthening the institutional and regulatory framework for the LRCP implementation in coordination with provincial and municipal bodies. The COMILU has a Consultative Committee to ensure the community participation and secure that COMILU takes into account the views of other public and private stakeholders from the basin region. The COMILU will be part of the Ministry of Infrastructure and Public Services of the PBA and will be part of the implementation of the project.

The initial COMILU organizational structure is as follows:



The Executive Directorate main role is :

- To provide assistance to the Board of the COMILU on matters regarding the implementation of directives and decisions adopted by the Committee.
- To support the strategic guidelines for the implementation of the Lujan Basin integral management.
- To provide strategic guidelines and formulate policies, plans, programs and projects on the matter of sustainable development of all economic and social activities involving natural resources and sanitation within the river basin, in coordination with the competent bodies.
- Outline policies to prevent and mitigate impacts, damages, or risks likely to arise on occasion of construction works in the Lujan River Basin.
- Outline education policies and raise awareness regarding the environmental issues of the Lujan River Basin, in coordination with the competent bodies.
- Promote mutual aid and cooperation agreements with provincial, national, government and non-government bodies aimed at formulating programs to preserve water resources in the Lujan River Basin.
- Design and implement actions to prevent and mitigate impacts and progressive reversion of critical scenarios in the Lujan River Basin, and play a role in the definition and design of indicators for environmental assessment of the Lujan River Basin, in coordination with the competent bodies.

- Establish appropriate intervention mechanisms to profit development opportunities of the Lujan River Basin and collaborate in the reversion of critical household situations in order to improve the quality of life of the Lujan River Basin population, in coordination with the competent bodies.
- Promote the environmental land-use planning of the Basin, addressing social, cultural, physical, economic, political, legal, and ecological aspects, coordinating common actions and policies with the Province, the Municipalities and other competent bodies.
- Monitor the implementation of plans, programs, projects, and actions relating to the Lujan River Basin, particularly those aimed at reversing or stopping environmental deterioration processes or environmental damages and those regarding prevention of and actions of response vis-à-vis environmental emergencies and catastrophes, in coordination with the competent bodies.

The Advisory Council :

- The Advisory Council of the Lujan River Basin Committee main role is to achieve the incorporation into the management of the Lujan River Basin of the views stated by organizations and people dealing with the Lujan River Basin. The members will be professionals or experts, on behalf of service users and intermediary entities; University representatives, training and research centers, and governmental and non-governmental organizations with agreements in force with the Lujan River Basin Committee (COMILU).

Basin Sanitation Planning Unit main responsibilities are:

- Undertake actions to improve the Basin Management Plan,
- Implement emerging programs and projects regarding integral Management of the basin's water resources.
- Design infrastructure programs, in coordination with the competent bodies.
- Consider the prevention of environmental and/or risk damages, and the mitigation and/or minimization of environmental impacts relating to actions and activities performed in the Lujan River Basin, in coordination with the competent bodies.
- Prepare and design projects for keeping duly updated minimal technical requirements for the Environmental Impact Assessment (EIA) of public or private works or activities conducted in the Basin, in coordination with the competent bodies.

Water Integral Management Unit main responsibilities are:

- Implement all policies adopted by the Board of the Lujan River Basin Committee, in coordination with the competent bodies.
- Manage the auditing, maintenance and exploitation of hydraulic works, drinking water supply systems and facilities to treat sewage effluents, in coordination with the competent bodies.
- Participate in activities and programs dealing with the environmental protection and conservation of the Lujan River Basin, in coordination with the competent bodies.
- Ensure the implementation of plans, programs, projects and actions within its area of competence, in particular, those directed towards reversing or stopping environmental deterioration processes or environmental damages and regarding the prevention of and actions of response vis-à-vis environmental emergencies and catastrophes, in coordination with the competent bodies.
- Detect harmful environmental effects likely to arise from industrial activities, services provision and public or private works developed in the Lujan River Basin, including actions to mitigate pollution effects, in coordination with municipalities and competent bodies.
- Control the pollution produced by liquid and gaseous effluents, solid and semisolid waste –special, pathogen, household and others– and manage the audits regarding compliance with standards set out under the legislation in force, in coordination with the competent bodies
- Collaborate in the prevention, surveillance and correction of environmental impacts, damages and/or risks relating to actions or activities performed in the Lujan River Basin, in coordination with the competent bodies.

- Collaborate in the environmental monitoring performed by the environmental police in the Basin, regarding waste producers, waste operators and waste processors –either special or pathogen– under the applicable laws, in coordination with the competent bodies.
- Implement surveillance or monitoring programs on environmental resources, in coordination with the competent bodies.
- Enforce the environmental terms of reference for implementation of projects bearing environmental impact, taking into account the preservation and conservation of the environment and the promotion of appropriate environmental technologies, in coordination with the competent bodies.

Management and Accounting Unit main responsibilities are:

- Accounting, finance and budget management of the unit.
- Manage, process and monitor procurement and tender processes originated in the COMILU Board.
- All human resources activities.
- All other administrative activities.

Institutional Relations Unit main responsibilities are:

- Advise the Board on the definition of policies and actions regarding information, education, and cooperation on sustainable development in the Lujan River Basin, in coordination with the competent bodies.
- Collaborate in the preparation and implementation of learning programs regarding environmental matters associated to the basin territory, in coordination with the competent bodies.
- Implement information and action policies in cooperation with non-governmental organizations and municipalities for the fulfillment of sustainable development objectives in the Lujan River Basin, in coordination with the competent bodies.
- Promote and implement mutual assistance agreements with international, governmental and non-governmental organizations to obtain technical cooperation and financial support for environmental matters falling under the remit of COMILU.
- Coordinate and implement environmental learning actions and campaigns referred to the Lujan River Basin, in coordination with the competent bodies.
- Implement education plans, programs and projects to raise awareness regarding environmental issues in the Basin, in coordination with the competent bodies.
- Implement actions to ensure environmental awareness among the population and involve citizens in the environmental protection of the Lujan River Basin, in coordination with the competent bodies.
- Disseminate, among the Lujan River Basin population all and any knowledge available on the natural and social environment of the Basin, to bring about attitude and behavior changes to achieve a better life quality, in cooperation with non-governmental organizations and competent bodies.
- Design and develop information systems to rely on a continuous diagnosis of the Lujan River Basin environmental conditions, and take necessary actions to inform the community.
- Implement a mechanism for community participation in sanitation and conservation plans and programs of the Basin and promote their articulation, in coordination with the competent bodies.
- Implement a system to meet the demands of the community regarding environmental issues of the Basin.
- Implement a system to meet the demands of public, national, provincial and municipal organizations regarding environmental issues of the Basin.

Legal Unit main responsibilities are:

- Advise the Board on every legal issue dealing with the enforcement and interpretation of regulations governing COMILU.
- Provide legal services, prepare legal opinions and reports, and draft administrative acts falling within its remit.
- Legal advice to the Board and the overall entity.

- Analyze, give an opinion and/or prepare draft-regulations deemed relevant by the Director.
- Provide assistance to the Board by assessing or issuing legal opinions on the administrative actions made by government or non-government entities or municipal, provincial, national or international organizations.
- Verify the enforcement of control mechanisms regarding the legality of administrative acts, which COMILU officials have adopted or may adopt in the exercise of the powers vested by the Board.
- Prepare legal opinions, draft administrative acts and complete any formalities regarding reports and/or clarifications requested from the Board by the judiciary, the legislative authorities and other provincial and municipal administrative jurisdictions.

Monitoring and Follow-Up

The monitoring and follow-up of the Project will be performed through a matrix of indicators including baselines and targets (expected results) and components (the outputs of the investment performed) with verification means as proposed. The Project budget includes resources for implementation of a monitoring system over activities and the assessment of these activities as necessary to verify compliance with performance indicators.

CAF Role:

CAF will play a significant role of control, both prior to and during the Project implementation. Such control will take place at all stages of the Project, to wit: prior to the first disbursement; prior to the tender process for each civil work and prior to the tender process of the Early Warning System (EWS), before the commencement of civil works and during disbursement periods which will require periodic reports - as established beforehand - as well as follow-up to environmental, social and other supervision aspects required for the adequate follow-up of the Project.



Please provide a project/programme implementation timetable in [section I \(Annexes\)](#). The table below is for illustrative purposes. If the table format below is used, please refer to the activities as numbered in Section H. In the case of outputs, please mark when all the required activities will be completed.

[illegible]

[illegible]

As per loan agreement with CAF, SFF will provide following reports:

- Semiannual. Progress reports submitted within forty-five (45) days after June 30 and December 31, each year.
- Annual report from the Project external auditor. Submitted within one hundred twenty (120) days following each fiscal year closing.
- Mid-term. When reaching fifty percent (50%) of disbursements or twenty-four (24) months counted as from the first disbursement, whichever occurs first.
- Final. Submitted within one hundred eighty (180) days following the latest disbursement.
- Other specific reports reasonably requested by CAF during the Project performance.

D.1. Value Added for GCF Involvement

PBA has included this Project among its investments priorities; however, there are budget and financial constraints. Those constraints are the reasons why PBA has divided the Project in two stages assuming the risk of delay in the commencement of stage 2 and the risk of failure in the full implementation of structural and non-structural measures.

GCF funding will allow full development of the Project removing the uncertainty of commencing stage 2 on time. The Project was conceived for an integrated management of the Lujan basin which will not be possible if it is not fully developed. Current problems derived from climate change have a root cause not only from the physical conditions of the basin but also from inadequate land- use, GCF funds, therefore, will provide the opportunity to implement measures for the adaptation to climate change in the early implementation of the project.

The GCF participation in the project is key to (i) ensure the integral implementation of the project encompassing both structural and non-structural measures such as institutional strengthening of the COMILU, the implementation of land-use plans and expropriations and resettlements derived from such measures. Furthermore, the availability of funds may allow starting the construction of areas for temporary retention of surplus water (ARTEH) in the upper and lower parts of the river basin and to substitute inflatable dams for lockgates at Mercedes and Lujan. The GCF resources ensures the implementation of non-structural measures that without the GCF participation would be at risk of not being incorporated, (ii) The Lujan River Basin Project is the first adaptation project in Argentina funded with reimbursable resources and incorporated in Argentina's budget and investment portfolio. This represents important changes in the usual approach of addressing climate change adaptation through non-reimbursable resources. This project seeks to combine public and climate long term investments that promote sustainable and climate-resilient development. The concessional financial terms and conditions of the GCF are essential to make a reimbursable operation for climate resilient infrastructure viable for the PBA.

GCF funding will provide the opportunity to rely on a financial structure in line with the timeframe for implementation of the Project and to witness timely economic, social and environmental results from the Project as expected. GCF funds bearing interest rates lower than current interest rates (0.75% *vis-à-vis* 9.125%), grace and repayment periods of 10 years (instead of 8 years) will enable PBA to invest future cash flows in the development of other priority projects. Today, the main PBA creditors are international bondholders, domestic bondholders and the national government (more than 90% of the debt). Involvement of multilateral organizations and other financial sources make up a low percentage of the whole PBA debt. CAF and GCF will enable PBA to reinforce the Project funding, improve the profile of its debt and liberate resources to fund other projects.

PBA will work together with GCF' and CAF' environmental and social experts, thus allowing institutional strengthening of PBA. The recently created COMILU will also benefit from the GCF and CAF know-how and expertise with long-term benefits in terms of sustainability and management of the river basin as well as better skills to be replicated in other priority basins in the Province.

The work on technical and environmental supervision will be enhanced with specialized external auditors and GCF involvement in environmental issues.

D.2. Exit Strategy

The Project has been conceived in such a way that non-structural measures ensure its future sustainability. This said, it is worth highlighting the following:

Creation and institutional strengthening of COMILU: This river basin committee was created by Law on May 2015. Its objective is to implement necessary actions for the preservation and sustainable integrated management of water resources. Its main functions are the following: formulate policies, strategic planning, programmes and projects on sustainable development of economic and social activities involving natural resources and human life in the river basin. Additionally, this Committee will also be responsible for the operational relationship with municipalities and civil society, and will coordinate contingency plans with civil defence or voluntary fire brigades in each jurisdiction. Unlike other river basin committees, this Committee has a management and participative role for the proper fulfilment of its objectives. PBA, from its yearly budget will allocate resources as necessary for funding future activities of the Committee (Law 14817 that created the COMILU):

Provincial Early Warning System (EWS): The Project should articulate with EWS. To this end the Directorate of Provincial Hydro-environmental Monitoring has been created to manage both the EWS and the Provincial Monitoring Centre to prevent and mitigate water contingencies. EWS implementation, further to institutional strengthening, will require both capacity building to civil stakeholders and the elaboration of evacuation plans. Early warning systems are a priority for Argentina and those are part of Argentina's commitments for adaptation to climate change; therefore, the future of this system will be assured if it is part of the Provincial Centre and will be part of the PBA budget.

Directorate of Provincial Maintenance. The competent authority for the operation and maintenance of Province hydraulic works is the SSIH through the Directorate of Provincial Maintenance. For the performance of its functions, the Directorate of Provincial Maintenance has the Directorate of Maintenance Supervision and the Directorate of Strengthening. The Directorate of Provincial Maintenance is responsible for the operation and maintenance of all projects financed with PBA budget.

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

E.1. Impact Potential

Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas

E.1.1. Mitigation / adaptation impact potential

The Project helps to fulfil the Fund's objectives of increasing river basin resilience to climate change. Flooding reduction translates into improving economic conditions for trade, agricultural production and industrial production, as well as improving conditions for people in the region. There will be a reduction of flooding damages, which will impact positively the people located in the river basin. Additionally, implementation of the Project will have a direct impact over 1.6 million people and indirectly over another 1.2 million people for a total of 2.8 million people, representing 17% of PBA population.

The Project will increase the adaptation capacity and reduce exposure to risks associated to climate change by means of two essential factors: a) implementation of an EWS to improve the population's prediction abilities and its capacity to react expeditiously and b) institutional strengthening. Such strengthening will secure better-trained teams duly qualified for the holistic management of the river basin, a civil society well trained on evacuation plans and duly skilled to construe and read information on early warnings. In addition, the interaction of different entities from the river basin under the leadership of COMILU will foster people awareness with respect to climate change, its effects and adaptation activities.

On the other hand, the Project financial mechanisms provide a new framework to be replicated in future adaptation efforts in Argentina. This would be the first adaptation project funded with public money in Argentina and the first project funded by GCF. GCF funding together with CAF funding, results in a sustainable and replicable financial mechanism.

E.1.2. Key impact potential indicator

Provide specific numerical values for the indicators below.

GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	
		Lifetime	
	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	Direct beneficiaries: 1.581.408 Indirect beneficiaries 1.214.240 Total beneficiaries: 2.795.648 ¹
		Percentage (%)	16.67% of the Buenos Aires Province (16.659.931 - total population)

¹ The calculation of direct and indirect beneficiaries was divided in:

Direct beneficiaries of the water conveyance works- enlargement of river channels- bridges enlargement and replacement- (Calculation based on Indec 2010 census and the area impacted by the works). Total: 314.199.
 Indirect beneficiaries of the water conveyance works- enlargement of river channels- bridges enlargement and replacement- (Calculation based on polygon and census radius). Total 1.110.404.
 Total beneficiaries impacted by the works: 1.424.603.

Beneficiaries of the Basin management and EWS: corresponds to the beneficiaries of the works and the rest of the projected inhabitants of the basin. For a total of: 2.795.648.

For maps a more detailed explanation please see Annex 13

Other relevant indicators	A key indicator to be considered is the avoided damages to dwelling derived from the implementation of the Project. Calculations include 6 scenarios of flood recurrence, considering the level of rainfalls, water height and damages to property and equipment, multiplied by the number of real properties affected, according to flooded surface, for each flooding scenario.				
	Recurrence (years)	Without project (USD)	With project (USD)	Damage avoided (USD)	
	2	87.937.831	35.577.984	52.359.847	60%
	5	106.208.246	43.278.805	62.929.440	59%
	10	116.520.213	49.583.578	66.936.635	57%
	25	130.708.645	58.186.311	72.522.334	55%
	50	145.953.798	63.111.719	82.842.078	57%
	100	165.687.864	71.428.936	94.258.929	57%
- Damages avoided under the most likely flood recurrence scenario (every 2 years with a probability of 50%) is of USD \$52.4 MM. (Ref- Lujan Basin study- Book III- ChapterIX).. - Hectares benefitted by the Basin management and the EarlyWarning System: 337.850. - Reduction of flood stains by 36% (6.879,12 has).					

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

The major change *vis-à-vis* other similar projects is that the Project at stake seeks to implement not only structural measures but also non-structural measures in accordance with the priorities for adaptation to climate change set out by Argentina in the INDC.

Improved Institutional Capacity – COMILU: River basins committees in Argentina are common, but the COMILU has been created with characteristics different from other committees. The scope and toolkit at COMILU disposal for the fulfilment of its objectives are the following:

- Management: The administrative structure will enable COMILU to perform an appropriate management.
- Institutional relationships: To better discharge its duties, COMILU will have representation in the municipalities of the river basin and will interact with members from the civil society, i.e., universities, civil defence, fire brigades, among others.
- COMILU will adopt measures for the environmental and territorial management such as delimitation of the riverbank, review and improvement of local land-use plans, determination of protected areas and, subsequent determination of required expropriations and resettlements.

COMILU operations may be an example to be replicated by other river basin committees in Argentina.

Improvement of Warning Systems: Implementation of the Early Warning System will enable the river basin to better react *vis-à-vis* climate changes and obtain better climate information compared to that available today. The EWS is part of Argentina's adaptation plans and its proper operation is crucial for replication in Argentina.

Improvement on Future Projects Approach: While certain past projects have focused on the construction of channels across the basin, such projects have been centered on water conveyance, rather than water retention. On the contrary,

the Lujan River Project is based on the development of areas for temporary retention of surplus water (ARTEH) and the substitution of inflatable dams for lock gates in Lujan and Mercedes. This approach may be replicated in other river basins seeking better effectiveness on adaptation measures.

Projects maintaining an appropriate ratio between benefits and the environmental impact of the works. The approach of this Project is to have an appropriate balance between the scope of civil works and conservation of natural key habitats such as floodplains, wetlands, reserve zones, etc. This balance should be replicated in future projects.

In order to be able to replicate the Project in other river basins, the following communication-related actions will be included: implementation experiences, implementation criteria and implementation models. These activities will be financed by the grant that is being requested to the GCF.

E.2.2. Potential for knowledge and learning

As mentioned in sections D.2 and E.2.1, the Lujan River Project will result in institutional strengthening of COMILU, the Provincial Directorate of Environment and the Provincial Directorate for Hydro-environmental Monitoring.

Further to said institutional strengthening, the Project provides agreements with the community, the civil society, the civil defence, and voluntary fire brigades in each jurisdiction. These stakeholders should also receive training in EWS and in evacuation plans. The objective of involving civil society in the Project (responsibility of COMILU) is to raise awareness on civil works and its adequate operation. This initiative responds to a growing demand and social pressure from the community and to higher requirements for environmental management and water resource preservation.

In terms of adaptation to climate change, this will be the first project implementing both structural and non-structural measures consistent with the priorities for adaptation to climate change set out in the INDC put forward by Argentina. These are practices to be replicated regionally.

From the financial point of view, projects in which regional entities (instead of national entities) are funded by multilateral organizations and GCF are a new alternative for countries in the region as such countries receive funds directly. Furthermore, a new source of financial resources with longer grace and repayment periods and lower financial costs improve the capacity of regional entities to implement projects on adaptation to climate change.

This Project may be an example to be replicated in other countries of the region with regard to climate change adaptation measures, development of integrated projects with structural and non-structural measures and development of projects in which regional entities are funded by multilateral organizations.

E.2.3. Contribution to the creation of an enabling environment

The Lujan River Project has been structured to implement joint structural and non-structural measures. With respect to non-structural measures, COMILU will be in close contact with the municipalities and the civil society. Implementation of the Project requires the conclusion of labour contracts (a function today entrusted to municipalities), and the improvement of land-use plans. Likewise, it is also required to coordinate contingency plans with defence civil entities and voluntary fire brigades in all jurisdictions and provide training to stakeholders. This working scheme calls for constant communication and interaction with the civil society and its involvement in environmental issues of the river basin, thus producing greater environmental awareness.

E.2.4. Contribution to regulatory framework and policies

In 1993 Argentina ratified the United Nations Framework Convention on Climate Change (UNFCCC) and in 2015 the Kyoto Protocol. Thus far, Argentina has submitted three communications under the United Nations Framework Convention describing Argentina's vulnerability to climate change, providing the inventory of gas emissions and advancing Argentina's position regarding adaptation measures, giving high priority to the implementation of such measures.

The following are Argentina's priorities under the United Nations Framework Convention on Climate Change:

- Introduce the expected climate change effects into parameters for designing infrastructure works, particularly, those related to water management and the construction of embankments to avoid flooding.
- Update laws and land-use plans taking into account expected climate changes, introducing statutes on land-use based on flood risk maps under current and forecasted climate conditions.
- Aim at having a land-use plans that take into account climate changes to avoid or mitigate undesired consequences over human settlements. This is relevant for developments along maritime coastlines and riverbanks.

Furthermore, prior to the Paris negotiations in 2015, Argentina, through the INDC, put forward its contributions on the matter of mitigation and adaptation measures making it clear that adaptation measures are an ongoing priority. Two out of the core seven adaptation measures to climate change are the following: a) intensifying and broadening early warning systems for heavy rains, floods and hot weather, and systems of response to and recovery from climate disasters; b) implementing structural and non-structural measures to address extreme events. Additionally Argentina's INDC calls for international financing to implement adaptation measures.

The Lujan River Basin has been structured taking into account Argentina's INDC and, therefore, is within Argentina's priorities for adaptation to climate change. Moreover, this is the first project on adaptation to climate change relying on international funding on the basis of reimbursable loans. The success of this Project would lay down the basis for structuring future projects on adaptation to climate change, both in Argentina and other countries in the region and worldwide, which have addressed the matter by means of non-reimbursable donations.

The Lujan River Project falls within Argentina's regulatory framework and PBA's field of competence. It is worth mentioning that Law 14.710, whereby the Committee of the Lujan River Basin is created, seeks to modify existing institutional arrangements enacted by the Water Authority more than a decade ago.

Law 14.710 vested COMILU with implementing powers for river basins management with legal authority to make and implement general interest decisions. Prior to the enactment of Law 14.710, Committees were of consultative nature to coordinate or arrange initiatives.

The purpose of Law 14.70 is the preservation of water resources and the sustainable management of the basin. To fulfil this objective COMILU has been vested with powers to formulate environmental policies of the river basin together with other competent organisms and powers for the integrated management of the river basin, including issuance of land-use plans and recommendations on related expropriations and resettlements.

Successful implementation of the Lujan River Project under Law 14.710 would constitute a regulatory framework replicable for projects on adaptation to climate change.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

Environmental, Social and economic co-benefits

Environmental Benefits

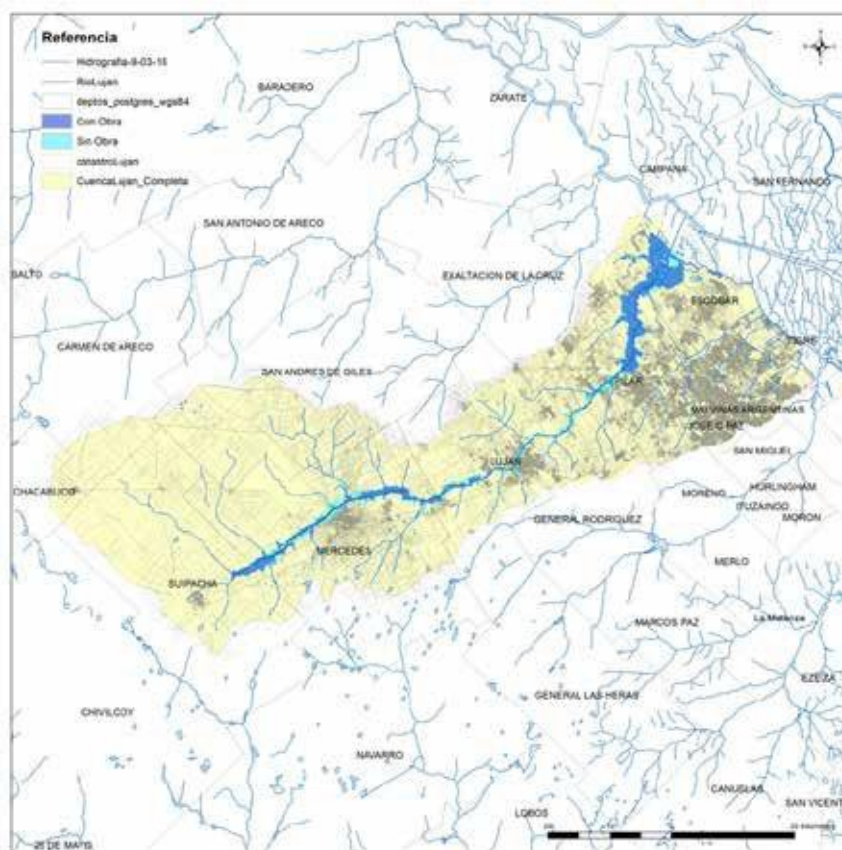
From the environmental point of view, two main benefits can principally be considered:

1. Hectares benefited by the Management of the Basin and the System of Monitoring and Early Warning

The hectares benefited by the management of the basin and the EWS are 337,850.74 which constitute 33% of the area of the plots parties involved and 1.1% of the total surface area of the PBA (30,490,700 hectares).

	Hectares
Campana	95.400,54
Escobar	30.300,75
Exaltación de la Cruz	63.400,17
General Rodríguez	36.000,14
José C. Paz	5.000,16
Malvinas Argentinas	6.300,09
Moreno	18.600,13
Pilar	38.300,01
San Fernando	87.700,08
Tigre	30.400,35
Chacabuco	228.700,38
Luján	77.700,13
Mercedes	104.900,47
San Andrés de Giles	113.200,04
Suipacha	94.300,87
Total	1.030.204,31

Source: INDEC, 2011. National Population, Households, and Housing 2010 - Total country and provinces - Definitive Results, Serie B N° 1.



2. Benefit of the work: flood stains with and without work .

On the other hand, from the analysis of the flood stains generated from the mathematical hydrological modelling in the current situation without works regarding the situation with works, the following results arise:

The current flood stains without works cover an area of 19,056.44 hectares, and in the circumstance with works, it is 12,177.31 hectares. This implies a reduction of 36%, represented by 6,879.12 hectares, with works executed that would be the hectares benefitted in a direct manner by the works.

It is important to consider that said reduction occurs spatially in the most populated areas, in the City of Mercedes and Lujan County, including the city's urban area and other locales with high social vulnerability such as Olivera, Villa Flandria (Figure 1) within Lujan, Manzanares, and Fatima counties among others in Pilar county (Figure 2).

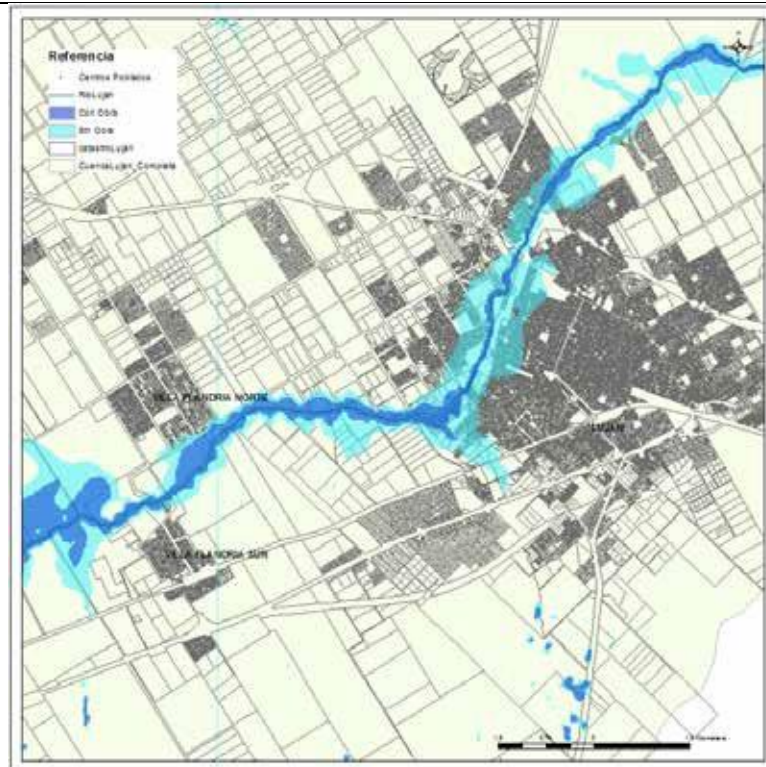
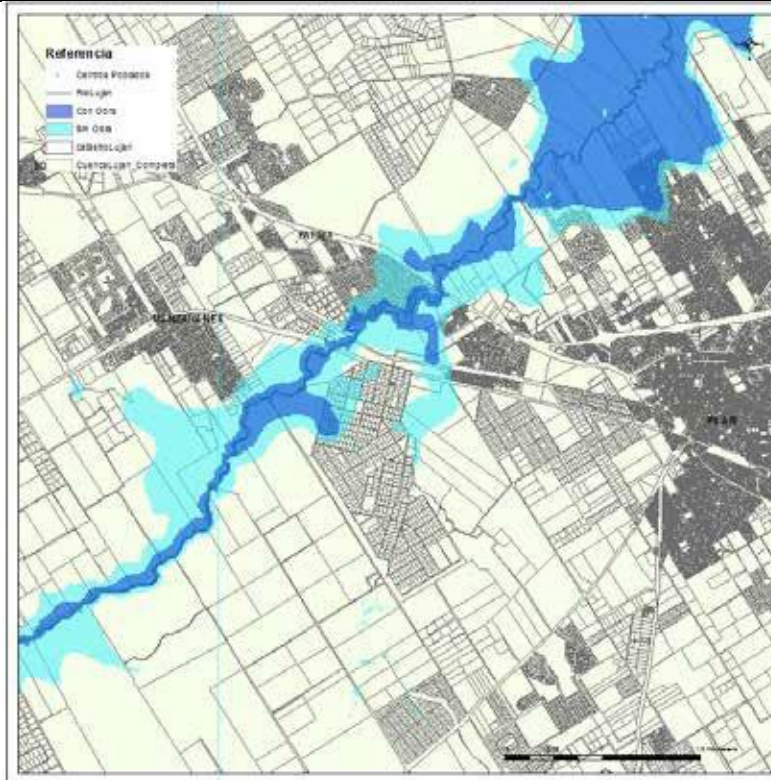


Figure 1: Localities of Villa Flandria North and South, and urban area of Lujan (Lujan County) flood stains with and without works.



Economic Benefits

The following are core tangible benefits:

- The most important economic benefit of the Project is to avoid the economic damages resulting from floods in areas inhabited by the population at risk when the Lujan River bursts its banks due to climate changes.
- The economic impact of the Project for a time horizon of 20 years has a cost/benefit ratio of 1.44; an economic net present value (ENPV) of USD \$116,91 MM and an economic internal rate of return (EIRR) of 20.2 %.

Overall, the implementation of the Project will bring about a reduction in damages to assets, services, urban infrastructure and the historic and architectural heritage of places affected by floods.

Social-Economic Benefits

Territorial management plans are focused on:

- The demarcation of the riverbank to: 1) define boundaries of private and public properties, and; 2) accurately identify flood-prone areas subject to administrative restrictions and easements.
- The overall review of municipal land-use plans.
- The creation of a network of areas protected within the river basin.

Those actions will minimize obstruction to surface runoff, avoid flood in private lands and enable biological preservation in protected areas, as requested by the civil society. This will imply an improvement of the quality of life of the population located in area of influence.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

Describe the scale and intensity of vulnerability of the country and beneficiary groups, and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc).

Overview of the Economic Conditions in Argentina

- Argentina's economy is in recession. According to the National Statistics and Census Institute (INDEC), the GDP of Argentina contracted by 0,7% in the first quarter of 2016.
- The accumulated inflation rate in the first quarter of 2016 was 30%.
- In 2015, the primary deficit was 1.9% of the GDP, the highest in the past 10 years.
- The new administration inaugurated in December 2015 put forward a development plan based on the following: free exchange market, monetary policy based on a target inflation rate, correction of the fiscal policy, investment and infrastructure plans and recovering access to financial international markets.
- A gradual recovery of the economy is expected based on the surge of foreign investment, consumption increases and access to foreign funds to fund the required infrastructure investments, that is, the driving force for economic recovery.
- During the new administration government investment in infrastructure is expected in the amount of USD \$33.225 MM. International financial organizations have committed USD \$9.000 MM (USD \$4.000 MM from international organizations) and, the remaining balance to be provided by the Argentinean Treasury by means of fiscal deficit.

The economic situation in Argentina is difficult and the recovery calls for foreign investment, international funding and domestic funding through fiscal deficit. With regard to the implementation of both environmental measures and adaptation measures, Argentina has always highlighted the need for international funding.

Economic Situation of PBA

- PBA is the most populated Province of Argentina with a population of 16.659.931 inhabitants (accounting for 39% of country's population). Encompasses the Great Buenos Aires (districts surrounding the city of Buenos Aires) the most densely populated area of the country.
- PBA gross domestic product accounts for 35,8% of Argentina's GDP and PBA industrial production accounts for 54,4% of total industrial production (according to the year average of 2009-2012).
- Capital investments have been low representing an average of only 4.4% of total expenses for the last 8 years. Such low capital investment resulted in deterioration of the road network, problems in the health and education systems, recurrent flooding and increasing insecurity.
- Therefore, the new administration is committed to:
 - Improve the road network, in particular, the access to ports.
 - Expand access to drinking water and to the sanitation network, particularly in the area of Great Buenos Aires
 - Increase resilience, reduce risks associated to flooding from Salado and Lujan rivers.
 - Improve the quality of schools and hospitals.
- Such investments require huge resources. The 61% of PBA income originates in the Province and the remaining 39% comes from the Federal government. The tax burden in the Province has increased in the past years and, therefore, there is little margin to increase such tax burden.
- After two years of balanced budgets, in 2015 there was a fiscal deficit. The public debt/GDP ratio remains below 7% and the debt is less than 50% of the total income of the Province.

Current Vulnerability of Argentina

Argentina is subject to high vulnerability vis-à-vis climate change. Despite the fact that Argentina ranks in the middle of the Index of Environmental Risk Exposure (World Bank, 2005), exposure to such risk increases because of the large number of glaciers and the large number of people living in low-lying coastal areas. In the past 30 to 40 years, Argentina has shown specific climatic characteristics in large part of its territory, most likely associated with climate changes. Those characteristics were the cause of certain type of disasters in sectors such as energy and agriculture. Broadly speaking, the most important trends are the following: (i) increase in the average annual rainfall in almost all the country, except in the mountain range area of the West and in some areas of the extreme Southern part of the country. This trend in the Northeast dates back from 1970s. The increase in rainfall ranged between 10% and 40% and was sharper in two areas of the country: West of Buenos Aires and North in the Pampa, and North-eastern parts of the country; (ii) increase in the frequency of heavy rains in the central and Eastern parts of the country. The frequency became higher since 1990s, and in some areas the frequency of heavy rains has tripled; (iii) increase in the average temperature in the Patagonian area since the second half of the twentieth century, reached temperatures of 1° C in the Southern part. These phenomena gave rise to an increase in the isotherm to 0° C and retreat of glaciers, sharp in 48 out of the 50 existing glaciers. The retreat is also observed in the Cumana zone.

The main consequences of this trend are the following: changes in water balance and hydrology across the central and North-eastern part of the country; increase in rainfall resulting in a westward shift of the isohyets corresponding to dry land farming; more frequent flooding of the Parana river giving rise to significant losses in the agricultural sector and the need to plan huge livestock evacuations, particularly in the Southern part of *Entre Ríos*; disruption of the land transport network – rural roads become impassable about 60 days a year – complicating farming and the delivery of products. In the North-eastern part of the country there was a generalized increase in the average annual rainfall and a greater internal variability, adversely affecting agriculture. In the region of Cumana, since 1980s the average riverstream flows have decreased between 50% and 60% (San Juan, de los Patos, Atuel, Mendoza and Tunuyán rivers), increasing the risk of a water shortfall. On the other hand, an increase in average temperature brings forward the process of snow fusion, increasing both streamflows and flooding risks and mudslides in spring. In the region of Comahue, during the last 20 years the average streamflows have decreased about 30% adversely affecting the generation of hydroelectric energy: the generation was 40% less than the generation in past climate conditions. The impact of retreating glaciers on streamflows is not yet well understood, but it may entail material losses of water reservoirs and other mechanisms on streamflow control.

According with Swiss Re (October, 2016)² over the past 50 years, 75 major flood events have been reported in the country, affecting around 13 million people and taking more than 500 lives. With the equivalent of USD 22.5bn lost since 1980, floods are the most costly natural catastrophe affecting the country. Adjusting for the country's GDP growth, the same events today would have caused accumulated losses of USD 43.5bn. Today 1 in 3 Argentinians live in areas highly exposed to flooding, amounting to a total of 14.2m people across the country. Most of the population living in flood-prone areas is concentrated in the Greater Buenos Aires, Pampas and Gran Chaco regions, which together account for 65% of highly exposed population. Looking ahead (flood-exposure in 2025), the impact of floods on the social and economic structure of Argentina is expected to worsen. Population growth, urbanisation, economic growth and climate change are the main drivers of this increase in flood risk exposure. Some options to close these gaps from the public sector are: structural and non structural measures. Structural measures for flood control Infrastructural measures have been at the core of the public sector flood management approach in the country. Building dams, dykes and catchment reservoirs can significantly reduce the damage caused by a flood event. Sound maintenance of existing infrastructure and periodic adaptation to reflect changes in flood exposure are also important. Temporary protection measures – e.g. controlling river flows with temporary sand bag walls and mobile flood barriers – can further reduce the risk. Non-structural measures for flood control Weather and hydrological models that enable forecasting and

² Staying afloat. Flood Risk in Argentina. Swiss Re. Expertise publication. October 6, 2016.
<http://www.swissre.com/library/expertise-publication/mantenerse-a-flote-el-riesgo-de-inundacion-en-argentina-long-version.html>

provide advance warning to regions at risk can be crucial to reduce property damage and save lives in case of severe events.

The following are expected climate changes in Argentina:

- Temperature rise: It is likely that in the period 2020-2040 temperature rises 1°C in Argentina. This temperature rise will be more significant in the North resulting in greater evaporation and, subsequently, in increases in both aridity and desertification.
- Changes in the pattern of rainfall: An increase in rainfall is expected in the central area of Argentina, mainly in summer and autumn. By contrast, a reduction in rainfall levels in the Andean region (Northeast of Patagonia and Comahue) is expected. Those changes in rainfall patterns are significant bearing in mind that 20% of the land area is already under water stress with considerable variations in the distribution of fresh water resources. Changes in rainfall and temperature increases translate into increased evaporation and reduced runoff (running water are not adequate for irrigation), and some areas become unsuitable for cropping. These phenomena adversely affect agriculture that accounts for a large percentage of exports from Argentina and also affect 54% of its land area. Agriculture should be positively affected assuming that fertilizing with CO₂ increases harvest of wheat (14%), maize (19%) and soja (67%) in the South whereas in the North a decrease is expected.
- Increase in the frequency of extreme climate events: more frequent rainfalls and floods are expected, thus adversely affecting the Northeast and central zones of the country, including Buenos Aires.
- Rise of the sea level: Most likely, average sea level will rise causing erosion in the coastal area, mainly in the Southern part of Patagonia. Buenos Aires may also be affected.
- Retreating of Patagonia and Comahue glaciers may continue: In 20 years the disappearance of glaciers in Patagonia is expected, thus adversely affecting consumption, agriculture and hydroelectric energy. In particular, it is estimated that hydroelectric energy generation in Provinces of Mendoza, Río Negro and Neuquén will be negatively affected by the downward trend of snows in the Andes. The hydroelectric generation in the region of Comahue, South of Argentina, that currently generates almost one fourth of total hydroelectric energy of Argentina, may be reduced by one third during the decade starting in 2020 because of decreases in river levels.
- Though indicators on human and social development have substantially improved after the 2002 crisis, levels of poverty and inequality in Argentina are similar to those prevailing during 1990s. There are significant differences between regions with high poverty rates, for instance rural Northeast and Northwest regions, where the economy is predominantly based on agriculture. These inequalities have significant impact with regard to the adaptation capacity of people.

Current Vulnerability of the Lujan River Basin

The Lujan River Basin has a population of approximately 2.8, accounting for 17.% of the population of the Province, with high concentration in urban areas. The population density in the region is 227,4 inhabitants per km², approximately four times higher than the population density of the Buenos Aires Province (50,8 inhabitants per km²)

Regarding levels of poverty, 11.4% households in the basin have at least one indicator of basic unsatisfied needs, a larger proportion than in the Buenos Aires Province (8.2%).

Across the river basin there is a mixed and varied land-use. Land-use in some regions is agriculture, manufacture and urban, whereas in others such land-use is basically urban.

The Lujan River Basin has a rich historic, archaeological and paleontological heritage, threatened by anthropogenic pressure, mainly urban pressure. This heritage is at risk.

E.4.2. Financial, economic, social and institutional needs

Financial resources

1. PBA needs financial resources to implement this Project and no new financial obligations should jeopardize PBA available cash flow. To that end, PBA should seek that the Project financial structure contains suitable terms and conditions.
2. Currently PBA's main source of financing is long-term bonds with an average maturity of 8 years and rate of interest of 9.125%. Those conditions are unsuitable for the Project.
3. Against that backdrop, the financial structure of the Project should be in accordance with PBA capacity of payment. To this end, the financial structure of the Project should:
 - a. Provide long repayment and grace periods and the debt maturity of the Project should not match the current PBA debt maturity.
 - b. Provide interest rates enabling to reduce funding costs.
 - c. Use new funding sources, as this is the first time PBA has structured a project on adaptation to climate change.

Institutional Strengthening:

The Project requires strengthening the recently created COMILU and the Provincial Directorate for Maintenance. These two bodies will be responsible for the future integrated river basin management and for the maintenance of works, respectively.

COMILU should have a highly skilled management team for the implementation of non-structural measures and should also have permanent interaction with all stakeholders in the river basin.

The Directorate of Provincial Maintenance should cover a large area and deal with a great variety of works and water bodies: channels, reservoirs, lakes, dams, floating lock gates, etc.

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

As mentioned in section C.1, in 1993, Argentina ratified the United Nations Framework Convention on Climate Change (UNFCCC) and the Kyoto Protocol in 2001. In 2002 it created the Secretariat of Environment and Natural Resources (SAYDS) under the authority of the Office of the Cabinet of Ministers. The SAYDS relies on the Directorate of Climate Change responsible for coordinating and implementing policies relating to with climate change.

Argentina has submitted three national communications to the UNFCCC to describe Argentina's vulnerability, the inventory on greenhouse gas emissions (GHG) and advancing its position on adaptation measures stressing that adaptation is a priority. Regarding priorities, Argentina intends to:

- Introduce the expected climate change effects into the parameters for design of infrastructure works, in particular those related with water management and construction of embankments to avoid floods.
- Update laws and land-use plans taking into account expected climate changes, introducing statutes on land-use based on flood risk maps under current and forecasted climate conditions.
- Aim at having a land-use plans that take into account climate changes to avoid or mitigate undesired consequences over human settlements. This is relevant for developments along maritime coastlines and riverbanks.

Furthermore, prior to the Paris negotiations in 2015, Argentina, through the INDC, put forward its contributions on the matter of mitigation and adaptation:

Mitigation: An unconditional commitment to reduce greenhouse gas emissions GHG (15% with respect to the “Business as Usual” scenario for year 2030) and a commitment conditional to obtaining international financing, technology transfer and capacity building (30% with respect to an scenario “Business as Usual” for year 2030).

Adaptation: This is Argentina’s main priority and for this reason it has been implementing adaptation measures. However, their expansion needs international funding, technology transfer and capacity building. Two out of the main seven adaptation measures considered in the INDC directly relate to the Lujan River Project:

- a. Intensifying and broadening early warning systems for heavy rains, floods and hot weather, and systems for response to and recovery from climate disasters.
- b. Implementing structural and non-structural measures to address extreme events.

The Lujan River Project is the first adaptation climate change project receiving public funds and reflecting Argentina’s commitment to comply with international commitments made on climate change adaptation. (No objection letter included in Annexes).

E.5.2. Capacity of accredited entities and executing entities to deliver

Buenos Aires Province:

As mentioned in section C4, PBA will implement the Project through the Subsecretariat of Finance from the Ministry of Economy and the Ministry of Infrastructure and Public Services (MISP) in PBA. The Province has extensive experience implementing projects and programmes funded by multilateral organizations. In the past ten years PBA government has concluded projects and programmes with international funding for an amount of over USD \$1.000 MM dealing with drinking water supply and sewage and drainage networks. PBA has ongoing projects for an amount of USD \$570 MM and over USD \$1.500 MM in the pipeline. In the last five years PBA has implemented an average of USD \$92 MM per year in water and sanitation projects.

On the other hand, despite being inexperienced entities, newly created institutions such as COMILU and the Directorate of Provincial Maintenance, are currently being provided with structures and mechanisms to discharge their duties and reinforce capacity building.

CAF:

In the past five and a half years CAF has disbursed 98 loans, approved USD \$4.652 MM and made disbursements for an amount of USD \$3.471 MM devoted to projects in Argentina, reflecting a high degree of experience in dealing with this kind of projects, as shown in the table below:

Table 1. Approvals, Disbursements and Total Debt – Argentina (USD \$MM)

	2011	2012	2013	2014	2015	2011 – 2015		2016*
						Total	Average	
Approvals	1.345	839	1.100	674	694	4.652	930	0
% of Total	13%	9%	9%	6%	6%	9%	9%	0
Disbursements	1.100	464	585	841	481	3.471	694	98
% of Total	9%	9%	8%	13%	8%	10%	10%	4%
Debt	2.438	2.682	2.922	3.269	3.215			3.085
% of Total	13%	13%	14%	14%	15%			14%

Furthermore, this project falls within CAF framework of strategic support to Argentina for the water sector. Currently, Provinces of the Great North are implementing sanitation projects with loans for an amount of USD \$240 MM and foresee to conclude a new loan for an additional amount of USD \$70 MM. On the other hand, the first phase of basic works on drinking water which implementing agency is AySA is being disbursed, and is pending the conclusion of a

new loan agreement for the second phase for an amount of USD \$240,5 MM. Finally, a new loan amounting to USD \$40 MM to develop a USD \$60 MM project for new irrigation areas is foreseen.

Regarding the Province of Buenos Aires, in 2015 CAF approved the Rehabilitation Program for PBA infrastructure for an amount of USD \$10 MM, whereas the Project of Construction of the Colorado River – Bahía Blanca Aqueduct, Phase I, for an amount of USD \$150 MM is being implemented.

Control mechanisms developed by CAF for its projects are to be applied to the Lujan River Project (Annex 2: Term Sheet CAF approval).

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

In the past years the PBA, within the framework of the Lujan River Basin Committee (COMILU), has held meetings with all districts integrating the Upper and Lower parts of the river basin.

Acknowledging the relevance of the dissemination of the Project at an early stages, the PBA conducted two presentations and preliminary consultations:

- (i) During the elaboration of the PMCRL. At that time, eight (8) meetings were held in five municipalities from February to May, 2015 and Advisory Commission of the Lujan River Basin. A summary of process was already presented to the GCF in the feedback sent on February 2, 2017.
- (ii) During the elaboration of the Risk Management Plan for the Lujan River Basin. From May to June, 2016 a series of three workshops were held to incorporate risk management strategies in the local context with emphasis on the floods in the basin. These workshops included technical site visits with municipal (Suipacha, Luján, Pilar, Campana, Exaltación de la Cruz, Escobar, San Andrés de Giles, Tigre and Mercedes), provincial authorities (Provincial Water Authority, OPDS, Land Planning, Security and Social Development) and technical staff as well as civil society organizations representatives.

In 2016, COMILU (created by Law 14.710) has held meetings in the municipalities of Lujan, Mercedes and Pilar with all districts (15) that are part of the Basin to thus disclose both the structural and non-structural measures of the Master Plan.

Additional stakeholder engagements have been performed by the PBA in August 2017 to inform and receive opinion of key stakeholders of the Lujan River Basin on the Project and its submission to the GCF. Meetings were held with representative from civil society, communities, and national, provincial and local governments.

During the Project execution, community relationship strategy will be developed for each work, identifying and prioritizing the particular social actors for each one, as well as determining their interests and concerns, building on the basis of initial stakeholder engagements initiated by the PBA, to present the Project to civil society and community representatives. Also, appropriate channels of communication, dissemination and participation will be established. The Community Relation Plan is included in the Environmental and Social Management Framework.

The National Directorate for Projects with International Lending Institutions of the Argentine Ministry of Finance and Economy, acting as National Designated Authority before Green Climate Fund, has ratified its commitment *vis-a-vis* the Project, both with regard to the Letter of Non-Objection and by means of discretionary approval to PBA's indebtedness, to thus structure this Project for adaptation to climate change, on innovative and reimbursable basis.

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

- Financial Efficiency**
 Access to this new funding source modifies the duration of PBA debt. Indeed, PBA debt is currently concentrated in international bonds agreed for 8-years (versus 20 years), an interest rate of 9.125 (versus 0.75%) and no period of grace (versus 5-year periods of grace). Feasibility of the Project is contingent upon access to GCF funds; otherwise it will be unfeasible. Moreover, the proposed Project financial structure is paramount for PBA sound financial structure.
 Implementation of the Project is contingent upon the combination of CAF, GCF and PBA resources. Currently the Project has approved USD \$ 100 MM from CAF and USD \$91.9 MM from PBA budget.
- Cost Efficiency**
 For the Project structuring, 8 alternatives were considered and their assessment took into account technical aspects of civil works, environmental conditions and economic and financial optimization. Economic assessment was based on benefits in terms of flood damages avoided in dwelling for a period of 20 years. Alternative 7 (deemed optimal) maximized the combination of the three assessment criteria mentioned above. This alternative has an ENPV of 116.91USD \$MM, an EIRR of 20.2% and a benefit/cost ratio of 1.44.
- Implementation Efficiency**
 Financial closing of the Project and, hence, availability of resources for its implementation are key elements for cost optimization ensuring efficiency in subsequent tender and procurement processes. On the other hand, by enforcing CAF procurement guidelines transparency will be guaranteed, as well as efficiency and competitive procurement processes.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

This being a Project for adaptation to climate change, this section does not apply.

E.6.3. Financial viability

- Access to GCF funds is paramount to have a sound financial structure for implementing the Project and to ensure that the Project achieves economic, social and environmental benefits within the agreed timeframe. Access to financial resources with interest rates lower than current interest rates (LIBOR+1% versus 9.125%), grace and repayment periods of 20 years (versus 8 years) will enable PBA to allocate future cash flows to the development of other priority projects. Today, the main PBA financial creditors are both international bondholders and the national government (more than 90% of the debt). Multinational financial organizations and others have minimal participation in the share of PBA debt. CAF and GCF would enhance PBA financing, improve PBA debt profile and help PBA to release resources to finance other needs.
- Economic benefits from the Project have been calculated in terms of flooding damages to dwelling, avoided by reason the Project. Such results were calculated for 6 scenarios of flood recurrence, considering the level of rainfalls, water height and damages to property and equipment, multiplied by the number of real properties affected, according to flooded surface, for each flooding scenario.

Recurrence (years)	Without project (USD)	With project (USD)	Damage avoided (USD)	
2	87.937.831	35.577.984	52.359.847	60%
5	106.208.246	43.278.805	62.929.440	59%
10	116.520.213	49.583.578	66.936.635	57%
25	130.708.645	58.186.311	72.522.334	55%
50	145.953.798	63.111.719	82.842.078	57%
100	165.687.864	71.428.936	94.258.929	57%

Damages avoided under the most likely flood recurrence scenario (every 2 years with a probability of 50%) is of USD \$52.4 MM. Therefore, the cost of delaying the Project implementation would bear costs as mentioned under

the 50% probability, that is, significant costs for PBA to further explain the reasons why civil society needs this Project.

E.6.4. Application of best practices

Unlike other projects of this kind, the Lujan River Comprehensive Plan sets forth the implementation of both structural and non-structural measures for an integrated management of the river basin, and it is not limited to the construction of infrastructure. This comprehensive plan enables the Project follow-up beyond 5 years contemplated for implementation of structural measures.

Non-structural measures provide implementation of EWS as part of Argentina's strategy for addressing adaptation to climate change, in addition to other benefits for the population in the river basin. EWS enable the collection of climate information and early reactions in face of extreme events.

Regarding structural measures, 8 alternatives were considered and their assessment took into account technical aspects of civil works, environmental conditions and economic and financial optimization. In other words, all relevant aspects of any project of this kind were taken into account to select the best possible alternative.

On the other hand, works for temporary retention of water surpluses (ARTEH), which allow controlling streamflows and reducing floods - resulting from the poor conveyance capacity of the river - were included among the structural measures. This is part of the Project best practices.

E.6.5. Key efficiency and effectiveness indicators

GCF core indicators	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)	
	(a) Total project financing	USD \$ _____
	(b) Requested GCF amount	USD \$ _____
	(c) Expected lifetime emission reductions overtime	_____ tCO ₂ eq
	(d) Estimated cost per tCO₂eq (d = a / c)	USD \$ _____ / tCO ₂ eq
	(e) Estimated GCF cost per tCO₂eq removed (e = b / c)	USD \$ _____ / tCO ₂ eq
The Lujan River Project being a Project for adaptation to climate change, this section does not apply		
Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)		
Describe the detailed methodology used for calculating the indicators above. Please describe how the indicator values compare to the appropriate benchmarks established in a comparable context.		
The Lujan River Project being a Project for adaptation to climate change, this section does not apply.		
Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)		

**** The information can be drawn from the project/programme appraisal document.**

F.1. Economic and Financial Analysis

The objective of the Project is to avoid economic damages to urban and rural infrastructure of the region resulting from floods when the Lujan river burst its banks. The economic proposal uses the methodology of economic cost benefit analysis to determine the Project feasibility for solving the environmental problems. The main benefits of investing in the Project are the prevention of substantial economic losses of property adversely affected by future floods, which according to estimates of the 2010 National Population Census and the 2004 National Economic Census may affect more than 5.000 single family dwellings, multifamily dwellings, industry and commerce premises. Additionally, resources will be invested to enhance early warning systems to continuously inform the community as to meteorological forecasts and adopt measures to anticipate potential events.

The economic parameters to estimate the benefit and the opportunity of implementing the Project are the following: ENPV, EIRR and Benefit/Cost ratio. The basis of the cost/benefit analysis is the following:

- **Estimate of Costs and Investments:** estimated costs shown below correspond to the implementation of civil works, design, supervision and control of the Project, as well as improved early warning systems for the prevention of potential natural events resulting from increased rainfall in the region. These alternatives consider the required investments for different levels of flood controls being alternative 8 the one with the largest investment (not necessary the best return). Investment estimates for each of the 7 alternatives of civil works (alternative 1 been Status- quo)– prices as of July 2015 – are the following:

1. Alternative 1 – *Status-quo* (without Project)
2. Alternative 2 – Enlargement of Bridges
3. Alternative 3 – Enlargement of Bridges + Channel Enlargement Minimum (-20%)
4. Alternative 4 – Enlargement of Bridges + Channel Enlargement Medium
5. Alternative 5 – Enlargement of Bridges + Channel Enlargement Maximum (+20%)
6. Alternative 6 – Enlargement of Bridges + Channel Enlargement Minimum + ARTEH
7. Alternative 7 – Enlargement of Bridges + Channel Enlargement Medium + ARTEH
8. Alternative 8 – Enlargement of Bridges + Channel Enlargement Maximum + ARTEH

Alternative	Investment at Market Prices (\$)	Investment at Economic Prices (\$)
Alternative 2	1.263.821.000	829.487.408
Alternative 3	3.733.283.733	2.450.277.251
Alternative 4	3.955.231.299	2.595.948.759
Alternative 5	4.678.631.834	3.070.740.390
Alternative 6	4.306.658.533	2.826.602.044
Alternative 7	4.498.606.099	2.952.583.563
Alternative 8	5.332.006.634	3.499.571.822

In addition, operative and maintenance costs were estimated as 4% of the initial annual investment.

- **Estimate Benefits:** The benefits of the Project have been calculated as theoretical damages avoided to dwellings and other property affected by floods under different scenarios of rainfall intensity. The different components of the analysis are the following:
 - **Theoretical Damage:** Damages suffered by property type for different heights of water level.

Average Theoretical Damage by Property Type				
Height (m)	0,2 - 0,4	0,4 - 0,8	0,8 - 1,2	>1,2
Single family dwellings	8.976	14.199	21.457	22.223
Multifamily dwellings	9.533	14.636	22.749	23.576
Commerce	2.684	3.216	5.096	5.670
Industry	15.022	21.027	27.713	33.803
Precarious dwellings	1.574	2.407	4.118	4.195

- **Flooded Area:** Based on Geographical Information Systems (GIS) a calculation was made as to the number of real properties flooded under different recurrences.

Number of units affected

Without Project						
Type of Property	RECURRENCE					
	2 Years	5 Years	10 Years	25 Years	50 Years	100 Years
Single family dwellings	4.198	5.018	5.488	6.276	6.909	7.560
Multifamily dwellings	143	196	215	282	320	353
Precarious dwellings	127	136	147	165	174	182
Industry	46	51	53	57	60	62
Commerce	285	311	328	348	364	377
TOTAL	4.799	5.712	6.231	7.128	7.827	8.534

- **Estimate of Avoided Damages:** Based on the number of real properties affected for each recurrence and damages by type of property, the value of flood damages was calculated under the scenario without Project. Damages avoided are the result of the difference between the scenario without Project and other alternatives with Project.

Recurrence (years)	Without project (USD \$)	With project (USD \$)	Damage avoided (USD \$)	
2	87.937.831	35.577.984	52.359.847	60%
5	106.208.246	43.278.805	62.929.440	59%
10	116.520.213	49.583.578	66.936.635	57%
25	130.708.645	58.186.311	72.522.334	55%
50	145.953.798	63.111.719	82.842.078	57%
100	165.687.864	71.428.936	94.258.929	57%

- **Expected Value of Benefits:** For each flooding scenario, the expected value of damages avoided, is calculated weighted by the amount corresponding to each flooding versus the probability of occurrence.

Expected Value

Damage Avoided	52.359.847	62.929.440	66.936.635	72.522.334	82.842.078	94.258.929	11.416.850	
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Probability	0,5	0,2	0,1	0,04	0,02	0,01	0,000	
Expected Value Damages Avoided	13.089.962	17.293.393	6.493.304	4.183.769	1.553.644	885.505	999.674	44.499.250

- **Increasing Value of Damages Avoided:** It is assumed that flood damages increase over time due to the following reasons:
- Increased number of dwellings and buildings in the flooding area.
 - Increased value of furniture and other fittings for accommodation.
 - Growth of business activities.

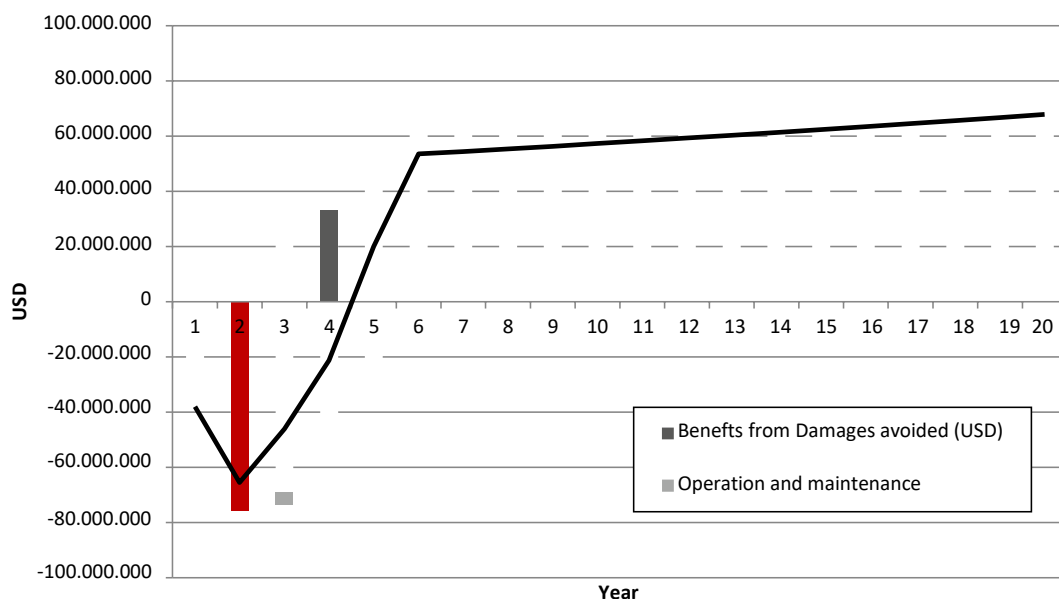
The Buenos Aires Province annual Gross Regional Product (GRP) growth rate of 2.003% was calculated by multiplying the estimated national GDP growth rate of 1.8% by the ratio registered in the period 1993-2013 (equal to 1.13). The coefficient obtained from the relationship between the total Gross Regional Product and the Gross Regional Product of commerce, services and industry ($R^2=0,97$ with statistical significance of 1% of the explicative variable "national GDP") was applied to the PBA annual GRP (2.003). The average growth rate was obtained by weighing each growth rate by the relative weight of participation.

Evolution of the GRP - Average Growth Rate ff Damage			
Type of damage	Share	Rate	Weighed Rate
Single family dwelling	90,2%	1,47%	1,3229
Multifamily dwelling	8,7%	1,54%	0,1336
Precarious dwelling	0,3%	2,08%	0,0067
Commerce	0,4%	1,96%	0,0001
Industry	0,4%	1,96%	0,0001
TOTAL	100,0%	1,47%	1,4634

The resulting average growth rate is 1,46%.

Alternative Economic Flows:

- The net present value of the economic flows was calculated by applying a discount rate of 12% for a time horizon of 20 years.



- Alternative 7 bears the largest net present value.

ENPV (USD \$MM)	116,91
EIRR	20,2%
B/C	1,44

In summary, Alternative 7 that defines the implementation of bears the largest net present value based on the fact that the structural measures to be implemented are enlargement of bridges + channel enlargement medium and ARTEH. These investments will reduce the flooding to a level that the value of damages avoided has a ENPV of 116,91 and and EIRR 20,2%.

Finally, to evaluate the economic impact of cost and benefit variations a sensitivity analysis was performed. The result shows that the Project is still economically acceptable.

Sensitivity Analysis				
Variable	Variation	EIRR	ENPV (USD \$)	B/C
Investment	10%	17,90%	90.440.987	1,31
	25%	15,00%	50.734.419	1,15
Benefits	-10%	17,66%	78.749.784	1,30
	-25%	13,61%	21.506.411	1,08
Combination	10%	15,49%	52.278.739	1,18

F.2. Technical Evaluation

The technological solution for the LRCP was based on the assessment of eight alternatives regarding technical aspects of civil works and both environmental and economic/financial optimization aspects. The optimal proposal took into account those three aspects.

The criteria used for the eight alternatives are the following:

1. Economic sustainability and promotion of capital generation.
2. Environmental sustainability.
3. Social benefits promotion.
4. Technical feasibility promotion.
5. Financial feasibility maximization.
6. Institutional and legal feasibility promotion.
7. Analysis of associated risks.

Bearing those criteria in mind, the range of options considered ranged between the *status-quo* and different channel enlargements, including (or not) works for retention of surplus water. The eight alternatives analyzed were:

9. Alternative 1 – *Status-quo* (without Project)
10. Alternative 2 – Enlargement of Bridges
11. Alternative 3 – Enlargement of Bridges + Channel Enlargement Minimum (-20%)
12. Alternative 4 – Enlargement of Bridges + Channel Enlargement Medium
13. Alternative 5 – Enlargement of Bridges + Channel Enlargement Maximum (+20%)
14. Alternative 6 – Enlargement of Bridges + Channel Enlargement Minimum + ARTEH
15. Alternative 7 – Enlargement of Bridges + Channel Enlargement Medium + ARTEH
16. Alternative 8 – Enlargement of Bridges + Channel Enlargement Maximum + ARTEH

The optimal alternative from a structural point of view should meet the following conditions:

1. A flexible solution adaptable to changes likely to arise during implementation of the Project (institutional or technological).
2. The development of civil works by stages to seek homogeneous and egalitarian improvements in different areas.

The optimal alternative from a financial point of view should meet the following conditions:

1. The possible implementations of the Project per stages taking into account BAP financial capacity to fully or partially implement the same.
2. The maximization of net economic benefits whereby the construction schedules can be based on productive possibilities in the region, by undertaking areas where better production possibilities exist.

From the environmental point of view, based on the fact that flooding are linked to a larger production of water due to climate change and/or soil response, and also to the growing incapacity to convey such volume of water, the optimal alternative should meet the following conditions:

1. Minimize flood-levels in large surrounding areas of the alluvial plain.
2. Increase river conveyance capacity in sections of the river with small slope.
3. Retain excess or surplus water where permitted by land conditions and minimize damages, if any, due to water accumulation.
4. Control peak streamflows, at least in case of small river flooding.

From the financial point of view the optimal solution should meet the following conditions:

1. The possible implementation of the Project per stages taking into account BAP financial capacity to fully or partially implement the same.
2. The maximization of net economic benefits whereby the construction schedule can be based on productive possibilities in the region, by undertaking areas where better production possibilities exist.

The alternative selected was alternative 7, which meets with construction and environmental requirements and bears the largest net present value.

F.3. Environmental, Social Assessment, including Gender Considerations

The *Management Plan of the Lujan River Basin Project* has been structured based on the “Study for the Lujan River Integrated Plan and Regulation and Sanitation Works Program”. Within the framework of the preparation of this study

consultations were held which served the Consultants and the Province of Buenos Aires, in order design the Project. Nonetheless, it is important to note that it is solely during the Project execution stage in which the final design of the works will be on hand, and which, as is the custom in bidding processes for the Province of Buenos Aires, will part of the participants' offer in each bid. In this stage, it will be required of the builders that, upon becoming contractors, they carry out consultation processes in those locales where the works will be implemented, in such a way as to complement those that the Province of Buenos Aires will do on the occasion of the elaboration of the Bidding Documents and their respective volumes of Technical, Environmental and Social Specifications (TESS).

The Lujan River Basin Management Project is of great importance for the Provincial Government of Buenos Aires and for the population of the Lujan Basin, especially for those that live or develop some economic activity in the flood zones and the opinion of the principal affected individual- which will be included through participative processes- which certainly will help those responsible for the final design of the works and the executors of those works, will establish the bases for a better operation and maintenance, for they will be works that the community will take ownership of.

The first item that will conform the Project's Environmental and Social Management Plan (ESMP), described in this document is the "Community Relations Plan", that will include spaces and methodologies in which, and through which, each person that feels affected, or that may want to be part of some work in particular, will be able to approach it before, during and after its construction period and lay out their opinions, desires, and observations, without regard to their slant.

Prior to beginning any work, the Contractor will have to possess a document in which the held consultations are explained and summarized and how the results of these consultations were included in the definitive design, were it the case (in case it were included, the reasons will have to be justified).

Afterwards, in a similar fashion, the contractor will have to look after establishing and maintaining active a channel of constant communication between each work and the Community, that permits the affected population or any interested party to be informed on the progress of the work, the progress of the implementation of those affected measures that were included or designed starting from the prior consultation or the establishment of a clear, limited, constant, fair and traceable mechanism for making complaints and resolving conflicts, including proposing one so that it be employed, after the necessary specific adjustments, for the Operator of said work.

The Community Relations Plan, from the start of the work, until its Operation and Maintenance stage, is the only thing that must be elaborated for any work that comes to form part of the Project to be financed, since the rest of the plans, depending on the work, on the area in which it is developed and on the environmental sensitivity of the staging site, among others, may be necessary or not.

In sum, the Environmental and Social Management Plan (ESMP), shall comprehend within its components, but without limiting itself to, the following aspects:

- a. Relations with the community (including a Stakeholder Engagement Plan; Grievance Mechanism, and Consultation Process)
- b. Compliance with (i) the applicable environmental legislation, (ii) of the conditions established in the authorization granted by the competent entity, (iii) Environmental and Social Safeguards of CAF, (iv) Safeguards of the GCF.
- c. Expropriations³ or Easements Plan, with a preponderance of this last option
- d. Prevention of water and soil contamination
- e. Preservation of air quality
- f. Control of erosion phenomena

³ The expropriation does not entail reallocation. A plot of land can be appropriated to establish a work without evicting any person.

- g. Restorations after the finalization of works
- h. Control in sites favorable for the development of illness vectors
- i. Maintenance of the quantity of surface water for human consumption and for irrigation, even if not expected to affect these aspects *a priori*
- j. Health and Safety
- k. Garbage and Solid Waste Management
- l. Preventive signage in work areas
- m. Preservation of fish fauna
- n. Preservation of archaeological and paleontological heritage
- o. Preservation of flora and fauna
- p. Unexpected Impacts Managements

It is the EIA [Environmental Impact Assessment] of each work that which will determine the preliminary ESMP on which later, with the incorporation of the conditions foreseen in the text of the authorization received from the Environmental Authority, the Provincial Bureau of Sustainable Development of the Province of Buenos Aires (OPDS, for its acronym in Spanish), and with the additions that the PBA or CAF, reasonably, may demand, the definitive ESMP will be set up on which, afterwards, the progress of environmental and social aspects will be compared.

Despite the fact that significant vulnerabilities have not been observed in the socioeconomic conditions of the population residing in the project's area of influence, the implementation of the project presents opportunities to incorporate women in various instances, which would facilitate their integration in the decision-making processes and would contribute indirectly to their political and economic autonomy.

In this sense, it is necessary that the processes of participation and public consultation which the COMILU will perform ensure an adequate and active participation of women, above all if it is held in account that the formal instances of consultation and participation of the citizens are limited⁹. Therefore, the Environmental Impact Study (EIS) must include consultation and citizen participation, with special attention to women, in accord with current regulations.

Below three entry points are presented for the incorporation of the gender perspective into the project:

- Environmental Impact Study. Analyzing the possibility of introducing gender considerations is recommended in this document, which could bring about activities within the Environmental and Social Management Plan
- Early Warning System. Studying the opportunity to incorporate the gender perspective in the design and execution of the system is recommended. Each group has distinct vulnerabilities in function to their culture, gender and other characteristics that affect in their capacity to prepare themselves effectively in the face of disasters, to prevent them and to respond to the same. The men and women perform different functions in the society and have a distinct level of access to information in disaster situations. Additionally, the elderly, disabled and socially and economically underprivileged persons tend to be the most vulnerable
- Institutional strengthening. Among the responsibilities of the COMILU is that of elaborating basin management instruments, which can be susceptible to incorporation of gender perspective.

As regards the operation's Gender Action Plan, the project will have a Project Operation Manual (POM), whose presentation is a requirement for the fulfillment of the conditions prior to the first disbursement.

For the scope of the project, it is held that the POM is the adequate tool for introducing the gender perspective. In this sense, working with the PBA to analyze feasibility of incorporating of these considerations and determining the appropriateness of elaborating a specific gender plan for the operation are recommended.

According with CAF's Environmental and Social Impact Report the project has moderate risk (category B, GCF – see the annex 7.2.).

F.4. Financial Management and Procurement

CAF funding requires - prior to the first disbursement - a Project Operative Manual detailing all operative processes and setting forth financial administrative aspects, including procurement processes.

In any case, the Project should comply with all CAF administrative and financial management standards, which are international standards. In this context, CAF's CPI and resolution state the following aspects:

- Prior to the first disbursement. Submitting, and copying Surety:
 1. Evidence that a bank account has been opened for loan funds to be deposited, and appointment of those responsible for managing such an account.
 2. Official appointment of a Project Coordinator as an interlocutor with CAF.
 3. Evidence that a Project Operational Manual (POM) has been drafted up including, at least: (i) an itemised definition of the Project (objectives, components, budget, and execution timetable; (ii) execution arrangements, including an organisation chart, duties and responsibilities, profiling of managerial and professional staff, including those related to the environmental and social management of the operation; (iii) a description of Project management coordination and oversight processes, including flowcharts; (iv) Terms of Reference ruling the Project's Environmental and Social Management Plan (ESMP), or the Project's Environmental and Social Specifications (ESS), in case this responsibility is transferred to contracting companies; (v) administrative and financial management; (vi) monitoring and assessment methodology including environmental and social issues; (vii) monitoring and reporting mechanisms; and (viii) the contents of biannual and yearly external mid-term and final auditing reports; among others.
- Over the disbursement period
 1. Prior to the 31st December of each year, submitting evidence that a suitable and independent entity has been hired to conducting an external audit of the Project, duly verifying on the basis of both, the Lending Agreement provisions, and the applicable local legislation, the contractual procedures being used, the compliance with the Lending contract clauses, and the use of both, loan resources and local counterpart resources. In the event that the first disbursement was to be made after the 1st of October, the contractual procedure could be taking place over the next calendar year.
 2. No later than ninety (90) days before the start of each calendar year, submitting the local counterpart's budget allocation corresponding to their investment in the project.
 3. At the time that each work, equipment or system being funded by the Project is provisionally taken in, an Operation and Maintenance Plan (OMP) should be submitted including: (i) a description of the operation, maintenance and management of the above-mentioned work, equipment or system; the names of those responsible; (ii) a quantification of resources required for operation and maintenance purposes, and financing sources; (iii) monitoring mechanisms; and (iv) agreement for transfer of works to a relevant entity, if applicable.
 4. At the time that fifty per cent (50%) of CAF loan disbursements have been reached, submitting a State-of-the-Art report dealing with the release of the trace or the space required for the completion of the temporary storage works as foreseen in Stage II of the Project.
 5. Upon reaching thirty percent (30%), sixty percent (60%) and ninety percent (90%) cumulative disbursements of the total loan, providing evidence that the timely provision of local counterpart resources to ensuring the normal execution of the project in its entirety is being duly abode by.
- Reports.

Reports as spelled out below should be submitted in agreement with the wording as spelled out in the MOP:

- Biannual. Progress reports, within the forty-five (45) days following the 30 June and the 31 December of each year.
- Project's Annual External Audit report. Within the one-hundred-and-twenty (120) days following the end of each fiscal year.

- Half-term report. Once fifty percent (50%) of disbursements or twenty-four (24) months after the first disbursement have elapsed, whichever comes first.
- Final. Within the one-hundred-and-eighty (180) days following the last disbursement.
- Other Reports. Specific reports that CAF may reasonably request over the Project's execution time.

As to financial control, the Project will have an external audit.

G.1. Risk Assessment Summary

Please provide a summary of main risk factors. Detailed description of risk factors and mitigation measures can be elaborated in G.2.

Six types of risks have been identified with respect to the Project:

1. The Basin Committee – COMILU – an organization recently created.
2. Implementation of new physical infrastructure and a monitoring system that must be operated and overhauled throughout the Project life.
3. Uncertainty regarding disposal of dredged material.
4. Unsound solid waste disposal.
5. Difficulties dealing with expropriations.
6. Problems arising in performed works.

G.2. Risk Factors and Mitigation Measures

Selected Risk Factor 1

Description	Risk Category	Level of Impact	Likelihood of Risk
<u>Committee of the Basin (COMILU) recently created:</u> The Committee of the Basin – COMILU – responsible for implementing non-structural measures, should: a) preserve water resources and ensure sustainable management of the Basin; b) ensure an integrated management of the Basin; c) plan land-use and proceed to expropriation or resettlements, as necessary; d) formulate environmental policies. Those actions should be undertaken in coordination with other agencies. The risk is the fact that COMILU is a recently created organization. COMILU should have a structural organization suitable for the Project and should be institutionally strong to implement any non-structural measures.	Social and environmental Social and environmental Social and environmental	High (>20% of project value) High (>20% of project value) High (>20% of project value)	

Mitigation Measure(s)			
1. The regulatory framework vests COMILU with powers for the integrated management of the Basin and for the implementation of activities pertaining to the Project. 2. The Project budget has allocated funds to ensure that COMILU is operative and runs under the required structural organization. 3. There is good coordination between Provincial and territorial, municipal and provincial agencies and entities. In addition, municipalities are represented in the board of directors of COMILU, ensuring COMILU coordination with other agencies. 4. PBA relies on highly skilled personnel and the Province is interested in a successful COMILU to replicate such experience in other river basins both at provincial and national level. 5. If the structure is not in place or not delivering as planned, the SSF should take charge of the COMILU, adjust the structure, secure its functioning and supervise it until results are achieved.			
Selected Risk Factor 2			
Description	Risk Category	Level of Impact	Likelihood of Risk
<u>Implementation of both physical infrastructure and monitoring systems that should be operated and be overhauled throughout the Project life:</u> The Project seeks to construct a new physical infrastructure and implement early warning systems that should be operated and be overhauled. Therefore, the allocation of human and economic resources is necessary to ensure proper functioning and sustainability of actions to be implemented.	Technical and Operational	Low (<5% of project value)	Low
Mitigation Measure(s)			
1. The Province recently created the Directorate for Provincial Overhauling responsible for overhauling water infrastructure in the Province. Despite of the fact that the Directorate has been recently created, PBA has highly skilled personnel to develop such actions. 2. Implementation of early warning systems is Argentina's priority under the framework of implementing adaptation measures to climate change, thus ensuring it's overhauling throughout the Project life. Furthermore, the Directorate of Provincial Hidro-environmental Monitoring will be responsible for the EWS, in addition to the management of other early warning systems in the Province. 3. The SSF is responsible for allocating the funds and ensuring implementation and functioning of the EWS.			
Selected Risk Factor 3			
Description	Risk Category	Level of Impact	Likelihood of Risk
<u>Uncertainty regarding disposal of material dredged.</u> Enlargement of the Lujan River channel will entail earthmoving and disposal of large volumes of soil. For the time being, no transportation routes or authorized disposal places have been identified.	Social and Environmental	Low (<5% of project value)	Low
Mitigation Measure(s)			
Prior the start of works, the implementing body will put forward a Plan for handling and disposal of material dredged.			

This risk will be mitigated CAF supervising mechanisms. CAF requirements prior disbursements and initiation of each work will trigger immediate actions with regards to compliance with licences. Implementation of actions will lie on the Ministry of Infrastructure and Public Services of the PBA (MISP).

Selected Risk Factor 4

Description	Risk Category	Level of Impact	Likelihood of Risk
<u>Unsound solid waste handling</u> Handling of household solid waste is inappropriate in the Lujan River Basin. Solid waste accumulates across the river basin contributing to the river overflow.	Social and Environmental Social and Environmental	Low (<5% of project value) Low (<5% of project value) Low (<5% of project value)	Medium

Mitigation Measure(s)

Prior the start of works, the Environmental and Social Management Plan will include a waste management plan.

Selected Risk Factor 5

Description	Risk Category	Level of Impact	Likelihood of Risk
<u>Difficulties dealing with expropriations and resettlements:</u> The Project may call for expropriations and resettlements; therefore, it exists the risk that some members of the community reject the Project.	Social and Environmental Social and Environmental	Medium (5.1-20% of project value) Medium (5.1-20% of project value) Medium (5.1-20% of project value)	

Mitigation Measure(s)

This risk may be mitigated through the elaboration of a detailed social plan, with permanent communication with the community, with permanent interrelation with municipalities and complying with federal and municipal regulations on expropriations and resettlements.

Selected Risk Factor 6

Description	Risk Category	Level of Impact	Likelihood of Risk
<u>Issues relating to performed works:</u> The risk that implemented works do not comply with technical specifications and, thus, fail achieving the objectives.	Technical and Operational Technical and Operational	High (>20% of project value) High (>20% of project value) High (>20% of project value)	

Mitigation Measure(s)

This risk may be mitigated with PBA and CAF supervising mechanisms. From the beginning of works there will be supervision ensuring proper implementation and compliance with technical specifications. Any action plan will have a follow up and this will be supervised by CAF and PBA control teams.

Other Potential Risks in the Horizon

* Please expand this sub-section when needed to address all potential material and relevant risks.

H.1. Logic Framework.

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level⁴

Paradigm shift objectives						
	<i>Describe how the proposed project/ programme’s expected contributions to global low-carbon and/or climate-resilient development pathways could be scaled-up and replicated including a description of the steps necessary to accomplish it.</i>					
	The Project will increase the resilience to climate change in the PBA by preventing flooding controlling stream-flows and attenuate the adverse effects from floods in fifteen municipalities of the Lujan River Basin.					
	By reducing floods, the project will improve economic conditions in the region, i.e. improved conditions for agricultural production, industrial production and commerce, as well as improved conditions for people in the region. Additionally, the implementation of the Project will mitigate damages in dwelling located in land with recurrent floods.					
	To achieve this objective, the project seeks solving problems in the river basin derived from physical and anthropogenic factors and, implement other non-structural measures.					
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Fund-level impacts						
A3.0 Increased resilience of Infrastructure and the built environment to climate change threats	3.1 Number and value of physical assets made more resilient to climate variability and change, considering human benefits.	Reduction of economic and human lives losses resulting as a consequence of flooding. This will be verified through the economic assessment post climate extreme events.	# Households: 6.723 Losses in 25 years USD \$127MM #Commercial outlets: 348 Losses in 25 years	Some damages prevented .	Damages avoided # Households: 3.470 Losses avoided in 25 years USD \$71.2MM #Commercial	Existing regulation is not fully supportive of Luján River project in relation to delimitation of floodlands, the expropriation and new settlements, the review and adaptation of new of local land- use plans as well as the definition of protected areas.

⁴ Information on the Fund's expected results and indicators can be found in its Performance Measurement Frameworks available at the following link (Please note that some indicators are under refinement): http://www.gcfund.org/fileadmin/00_customer/documents/Operations/5.3_Initial_PMF.pdf

			USD \$1.9MM		outlets: 101	
			Industrial outlets: 57		Losses avoided in 25 years USD \$ 0.7MM	
			Losses in 25 years USD \$1.8MM		Industrial outlets: 17	
					Losses avoided in 25 years USD \$ 0.7MM	

H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/programme outcomes	Outcomes that contribute to Fund-level impacts					
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	7.2 Number of males and females reached by (or total geographic coverage of) climate related early warning systems and other risk reduction measures established/strengthened	Reduction of economic and human lives losses resulting as a consequence of flooding. This will be verified through the economic assessment post climate extreme events Gender inclusion in	Losses in 25 years USD \$130.1MM	Some damages prevented.	Damages avoided in 25 years USD \$72.6MM	Lujan Basin population is receptive to the EWS and reacts to its warning as instructed.

		tenders, measures and plans. This will be verified in the tender documents.				
A5.0 Strengthened institutional and regulatory systems for climate-responsive planning and development	5.1 Institutional and regulatory systems that improve incentives for climate resilience and their effective implementation	i) Alerts issued by EWS ii) Independent assessment of the COMILU functioning and institutional strengthening	To date, the Basin has no management powers or authority to set out environmental policies	i) The process for the river basin management plan have been initiated. ii) 50% of the procurement and installation of the EWS.	i) The management plan for the river basin has been adopted. ii) EWS in place and COMILU operation and functional iii) EWS has gender-sensitive plans	Lujan Basin population adopts all the regulatory changes implemented by COMILU.
A5.0 Strengthened institutional and regulatory systems for climate-responsive planning and development	5.2 Number of gender-friendly policies, institutions, coordination mechanisms and regulatory framework that improve incentives for climate resilience and their effective implementation.	i) Review of the works operation manuals to assess the inclusion of gender sensitive measures ii) Tender documents for EWS and works include gender-sensitive requirement iii) Number of tender documents written in a gender-	To date, the Basin doesn't have gender-sensitive plans nor policies	i) Operative manual include gender-sensitive measures ii) Tender documents written in a gender – sensitive manner iii) Women and men attended consultations	i) Operative manual include gender-sensitive measures ii) Tender documents written in a gender – sensitive manner iii) Women and men attended consultations	The project had a gender – sensitive orientation in its implementation.

		inclusive manner iv) Minutes, list of participants and reports from public consultation demonstrate the participation of women and men.				
Project/programme outputs	Outputs that contribute to outcomes					
1. Engineering studies elaborated for tender documents	Number of engineering studies for tender documents All tender documents for the development of these works written in a gender inclusive manner	Executive projects and tender documents	Master plan of the Lujan River Basin	5 engineering studies for tender documents	5 engineering studies for tender documents	These studies are included in the budget:
2. Water conveyance works implemented and river channel enlarged	Number of works executed for the Improvement of the conveyance capacity of the Lujan River through construction of supplemental channels, enlargement of channels and crossing works, wide	Reports on technical inspection by the Provincial Directorate for Hydraulic Works	0	N/A	2 Inflatable dams; 5 enlargements of river channels.	a) These works are included in the Project budget. b) Works are performed in line with every technical specification and in due time. c) Work performed was gender inclusive.
3. Areas for temporary retention of surplus water (ARTEH) and	Number of Areas of ARTEH and substitution	Reports on technical inspection by	0	N/A	7 ARTEH installed with	a) These ARTEH works are included in

replacement of lock gates implemented	of inflatable dams for locked gates installed.	the Provincial Directorate for Hydraulic Works			inflatable dams for locked gates.	the Project Budget. b) Works are performed in line with every technical specification and in due time. b) Works are performed in line with every technical specification and in due time. c) Work performed was gender inclusive.
4. Bridges replaced and enlarged.	Number of bridges demolished, enlarged and cleared of 3 bridges that will improve current runoff conditions, removing river obstructions and detritus	Reports from the independent third-party evaluation Partial reception of the works Final reception of the certificate on compliance by the MISP.	Bridges in their current stage prior works	N/A	5 bridges demolished; 7 bridges enlarged; 3 bridges cleared.	a) These works are included in the Project budget. b) Works are performed in line with every technical specification and in due time. c) Work performed was gender inclusive.
5. Environmental and land use issues resolved	i) Policies defining adequate land use ii) Number of workshops to gather and diffuse lessons learnt iii) Documents and communication	i) Policies adopted ii) List of participants, minutes and reports from the workshops iii) Documents published	50% advancement on the river bank definition	100% of River bank lines defined. 4 workshops organized Report on initial lessons learnt.	River bank line defined. Protected Areas defined. Local land-use plans reviewed and found appropriate Protection	

	tools to gather and diffuse lessons learnt from the project				areas defined. 12 workshops organized. Reports on lessons learnt	
6. Early Warning Systems (EWS) developed and implemented	Functioning EWA	a) EWS approved by DPOH and integrated to Provincial Early Warning Systems b) alerts generated by the EWS	a) No Early Warning System available	a) 50% of the procurement and installation of the EWS.	a) Monitoring and surveillance systems in due operative conditions. b) Alarm and alert systems in due operative conditions. c) Communication systems in due operative conditions. d) Evacuation systems in due operative conditions. e) All outputs should be built in a gender-responsive manner.	a) Lujan Basin Population adopts the EWS and reacts to its warnings as planned. b) Lujan population participates in the EWS trainings.
7 Strengthened COMILU	Functioning COMILU	Minutes/ reports of the meetings held by the committee	Recently issued law creating the committee	i) The process for the definition of the river basin manageme	COMILU performed its role to the integral manageme	COMILU regulatory changes are adopted by the Lujan River

		Independent assessment of the COMILU functioning		nt plan have been initiated. ii) mid term assessment report of the COMILU functioning	nt of the Basin The Provincial Directorate for Maintenance has achieved proper overhauling of civil works.	basin population. Population actively participates in the project consultations-.
Activities	Description	Inputs		Description		
1.1. The Project should adopt an Operative Manual with gender consideration	Adoption of an Operative Manual outlining the Project operations and performance	1.1.1. Project definition. 1.1.2 Operations structure. 1.1.3 Terms of reference relating to the Environmental and Social Management Plan.		At the outset, it is required to define the Project operations to thus ensure its adequate performance taking into account gender-sensitive measures.		
1.2. Hiring of engineering studies	Third party hiring to carry out engineering studies and thus place the Project at feasibility levels.	1.2.1. Specifications on the scope of such studies. 1.2.2 Tender documents to hire the relevant studies.		PBA is required to define the scope of engineering studies as an input to draft tender documents relating to engineering studies. Bidding of the relevant studies is made through tender documents as defined, as well as the completion of contracts entered to carry out the relevant studies.		
2.1. Contracting of works aimed improving water conveyance and enlarging the river channels.	Third party hiring to perform civil works. Documents need to be written in a gender- inclusive manner:	2.1.1. Engineering specifications. 2.1.2 Tender documents written in a gender- sensitive manner.		PBA is required to define all engineering specifications as an input to draft tender documents. Bidding of the relevant civil works is made through tender documents as defined, as well as the execution of contracts awarded.		
2.2. Obtaining permits and licenses for the civil works	Obtention of permits and licenses required under Law is mandatory prior to the commencement of civil works	2.2.1. Applicable laws 2.2.2 Definition of works to be performed		PBA and the concerned third party is required to process all and any licenses and permits prior to the commencement of civil works.		

2.3. Plans on Environmental Management	Implementation of plans on environmental management relating to the civil works. Plans need to include gender-sensitive mechanism for public participation and consultations.	2.3.1. LRCP environmental study 2.3.2 Definition of technical specifications for the civil works	Rely on the approval of plans on environmental management prior to the performance of civil works. Consultation processes included men and women attendance.
2.4. Works aimed at improving water conveyance and enlarging the river channel.	Contract with third party hired to perform civil works:	2.4.1. Engineering specifications. 2.4.2 Tender documents written in a gender- sensitive manner. 2.4.3 Signed contract with third party.	PBA is required to sign a contract with the chosen third party and to comply with the tender documents and the technical specifications previously defined.
2.5. Works reception. Water conveyance.	Provincial Directorate for Hydraulic works performs technical inspection	2.5.1. Engineering specifications. 2.5.2 Tender documents written in a gender- sensitive manner. 2.5.3 Signed contract with third party. 2.5.4 Physical works.	PBA is required sign a formal document whereby it agrees that works performed did comply with technical specifications, tender documents and signed contract.
3.1. Contracting of works for temporary retention of surplus water and replacement of lock gates implementation-	Third party hiring to perform civil works. Documents need to be written in a gender- inclusive manner:	3.1.1. Engineering specifications. 3.1.2 Tender documents written in a gender- sensitive manner.	PBA is required to define all engineering specifications as an input to draft tender documents. Bidding of the relevant civil works is made through tender documents as defined, as well as the execution of contracts awarded.
3.2. Obtaining permits and licenses	Obtention of permits and licenses required under Law is mandatory prior to the commencement of works	3.2.1. Applicable laws 3.2.2 Definition of works to be performed	PBA and the concerned third party is required to process all and any licenses and permits prior to the commencement of civil works.

3.3. Plans on Environmental Management	Implementation of plans on environmental management relating to the civil works. Plans need to include gender-sensitive mechanism for public participation and consultations.	3.3.1. LRCP environmental study 3.3.2 Definition of technical specifications for the civil works	Rely on the approval of plans on environmental management prior to the performance of civil works. Consultation processes included men and women attendance.
3.4. Works aimed building temporary retention surplus water and replacement of lock gates implemented	Contract with third party hired to perform civil works.	3.4.1. Engineering specifications. 3.4.2 Tender documents written in a gender- sensitive manner. 3.4.3 Signed contract with third party.	PBA is required to sign a contract with the chosen third party and to comply with the tender documents and the technical specifications previously defined.
3.5. Works reception. Temporary retention surplus water and replacement of lock gates.	Provincial Directorate performs technical inspection	3.5.1. Engineering specifications. 3.5.2 Tender documents written in a gender- sensitive manner. 3.5.3 Signed contract with third party. 3.5.4 Physical works.	PBA is required sign a formal document whereby it agrees that works performed did comply with technical specifications, tender documents and signed contract.
4.1. Contracting of works for bridges replacement and enlargement-	Third party hiring to perform civil works. Documents need to be written in a gender- inclusive manner:	4.1.1. Engineering specifications. 4.1.2 Tender documents written in a gender- sensitive manner.	PBA is required to define all engineering specifications as an input to draft tender documents. Bidding of the relevant civil works is made through tender documents as defined, as well as the execution of contracts awarded.
4.2. Obtaining permits and licenses	Obtention of permits and licenses required under Law is mandatory prior to the commencement of works	4.2.1. Applicable laws 4.2.2 Definition of works to be performed	PBA and the concerned third party is required to process all and any licenses and permits prior to the commencement of civil works.

4.3. Plans on Environmental Management	Implementation of plans on environmental management relating to the civil works. Plans need to include gender-sensitive mechanism for public participation and consultations.	4.3.1. LRCP environmental study 4.3.2 Definition of technical specifications for the civil works	Rely on the approval of plans on environmental management prior to the performance of civil works. Consultation processes included men and women attendance.
4.4. Works for bridges replacement and enlargement implemented	Contract with third party hired to perform civil works.	4.4.1. Engineering specifications. 4.4.2 Tender documents written in a gender- sensitive manner. 4.4.3 Signed contract with third party.	PBA is required to sign a contract with the chosen third party and to comply with the tender documents and the technical specifications previously defined.
4.5. Works reception. Bridges replacement and enlargement-.	Provincial Directorate performs technical inspection	3.5.1. Engineering specifications. 3.5.2 Tender documents written in a gender- sensitive manner. 3.5.3 Signed contract with third party. 3.5.4 Physical works.	PBA is required sign a formal document whereby it agrees that works performed did comply with technical specifications, tender documents and signed contract.
5.1 Definition of river bank lines	COMILU must coordinate the definition of river bank lines, together with PBA and the concerned municipalities	5.1.1 Updated inventory of lands 5.1.2 Feasibility studies concluded	River bank lines will be defined on the grounds of the updated inventory of lands and concluded feasibility studies
5.2 Local land-use plans reviewed and appropriate	COMILU must coordinate the definition of land-use plans, together with PBA and the concerned municipalities.	5.2.1 River bank lines defined 5.2.2 Updated land-use plans 5.2.3 Land surveys	Update new land-use plans to thus define protected areas.
5.3 Protected areas defined	COMILU must coordinate the identification of protected areas, together with PBA.	5.3.1 River bank lines defined	River bank lines defined and new land-use plans required to identify protected areas and

		5.3.2 Updated land-use plans	expropriations to be completed.
5.4 Expropriations performed.	COMILU must carry out expropriations according to the definition of land-use plans and protected areas.	5.3.1 River bank lines defined 5.3.2 Updated land-use plans 5.3.3 Protected areas	Complete any expropriations on the basis of protected areas.
5.4 Resettlements.	COMILU must conduct any resettlement arisen from expropriations	5.3.1 Expropriations plan 5.3.2 Resettlement plan.	Resettlements derived from expropriations as defined in the plan must be concluded.
6.1. Hiring early warning systems (EWS)	Hiring third parties to provide EWS and its implementation. EWS built in a gender – responsive manner.	5.1.1. Technical specifications of the required EWS. 5.1.2 Tender documents written in a gender- sensitive manner.	PBA must identify technical specifications to hire EWS as an input to draft tender documents. Complete the bidding EWS providers and formalize the relevant contracts on the grounds of tender documents defined.
6.2. Works aimed at implementing an EWS executed	Contract with third party hired to perform implement EWS	6.2.1. Technical specifications of the required EWS 6.2.2 Tender documents written in a gender- sensitive manner. 6.2.3 Signed contract with third party.	PBA is required to sign a contract with the chosen third party and to comply with the tender documents and the technical specifications previously defined.
6.3. Works reception of EWS	Provincial DPOH performs technical inspection	6.5.1. Engineering specifications. 6.5.2 Tender documents written in a gender- sensitive manner. 6.5.3 Signed contract with third party. 6.5.4 Physical and technical works.	PBA is required sign a formal document whereby it agrees that works performed did comply with technical specifications, tender documents and signed contract.

7.1 COMILU Regulations	Draft the COMILU regulations taking into account gender considerations.	7.1.1 Creation of COMILU	Determine regulations applicable to the Basin Committee, to thus ensure its proper operations.
7.2 Start-up the COMILU	Make the COMILU fully operative	7.2.1 COMILU Regulations 7.2.2 COMILU organizational structure 7.2.3. COMILU budget as approved 7.2.4 Definition of COMILU processes and procedures	Such input, together with the hiring of personnel required will be the basis to start-up the COMILU.



H. Arrangements for monitoring, reporting and evaluation

Besides the arrangements (e.g. semi-annual performance reports) laid out in AMA, please provide project/programme specific institutional setting and implementation arrangements for monitoring and reporting and evaluation. Please indicate how the interim/mid-term and final evaluations will be organized, including the timing.

CAF funding requires certain reports on the relevant stages:

Reports. Submission of the following reports, according to the contents set out in the MOP:

- *Semiannual.* Progress reports submitted within forty-five (45) days after June 30 and December 31, each year.
- *Annual report from the Project external auditor.* Submitted within one hundred twenty (120) days following each fiscal year closing.
- *Mid-term.* When reaching fifty percent (50%) of disbursements or twenty-four (24) months counted as from the first disbursement, whichever occurs first.
- *Final.* Submitted within one hundred eighty (180) days following the latest disbursement.

Other. Specific reports reasonably requested by CAF during the Project performance

I. Supporting Documents for Funding Proposal

- X NDA No-objection Letter
- X Feasibility Study
- X Integrated Financial Model that provides sensitivity analysis of critical elements (xls format, if applicable)
- X Confirmation letter or letter of commitment for co-financing commitment (If applicable)
- X Project/Programme Confirmation/Term Sheet (including cost/budget breakdown, disbursement schedule, etc.)
- X Environmental and Social Impact Assessment (ESIA) or Environmental and Social Management Plan (If applicable) – Environmental and Social Report included in the annexes and post it at CAF's website
- X Appraisal Report or Due Diligence Report with recommendations (If applicable)
- ☐ Evaluation Report of the baseline project (If applicable)
- X Map indicating the location of the project/programme
- X Timetable of project/ programme implementation



República Argentina - Poder Ejecutivo Nacional
2016 - Año del Bicentenario de la Declaración de la Independencia Nacional

Nota

Número: NO-2016-02633522-APN-DNPOIC#MH

CIUDAD DE BUENOS AIRES
Miércoles 26 de Octubre de 2016

Referencia: Envío de Carta Compromiso para el Fondo Verde - "Project for the Management Plan of the Lujan River Basin"

A: Ruben Ramirez (CAF).

Con Copia A:

De mi mayor consideración:

Se remite en adjunto la Carta Compromiso firmada en original al Fondo Verde para la financiación parcial del Proyecto "Project for the Management Plan of the Lujan River Basin".

Sobre el particular, se solicita tenga a bien realizar las gestiones que estime pertinentes.

Sin otro particular saluda atte.

Digitally signed by AGUSTIN DANIEL MAI
DN: cn=GESTION DOCUMENTAL ELECTRONICA - GDE,
ou=ADMINISTRATIVA, serialNumber=CUIT 30715117264
Date: 2016.10.26 19:17:56 -03'00'

Agustin Mai
Director Nacional
Dirección Nacional de Proyectos con Organismos Internacionales de Crédito
Ministerio de Hacienda y Finanzas Públicas

Lic. Agustín Daniel Mai
A/C Director Nacional de Proyectos con
Organismos Internacionales de Crédito

Digitally signed by GDE
DN: cn=GESTION DOCUMENTAL ELECTRONICA - GDE,
ou=MINISTERIO DE MODERNIZACION, ou=SECRETARIA DE
MODERNIZACION ADMINISTRATIVA, serialNumber=CUIT
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Date: 2016.10.26 19:17:56 -03'00'



Ministerio de Hacienda y Finanzas Públicas

To: The Green Climate Fund ("GCF")

Buenos Aires, 10/26/2016

Re: Funding proposal for the GCF by CAF - Banco de Desarrollo de América Latina regarding Implementation "Project for the Management Plan of the Lujan River Basin"

Dear Executive Director Javier Manzanares,

We refer to the project "Project for the Management Plan of the Lujan River Basin" in Argentina as included in the funding proposal submitted by CAF – Banco de Desarrollo de América Latina to us on September 26th, 2016.

The undersigned is the duly authorized representative of Agustin Mai, the National Designated Authority/focal point of Argentina.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Argentina has no-objection to the project as included in the funding proposal;
- (b) The project as included in the funding proposal is in conformity with Argentina's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Name: Lic. Agustín Mai

Title: Director Nacional de Proyectos con Organismos Internacionales de Crédito.
Ministerio de Hacienda Y Finanzas Pública de la República Argentina.

Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Implementation Project for the Management Plan of the Lujan River Basin
Accredited entity	CAF
Environmental and social safeguards (ESS) category	Category B
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2017-08-30
Language(s) of disclosure	Spanish and English translation
Link to disclosure	https://www.caf.com/es/temas/a/ambiente-y-cambio-climatico/proyectos/#itemproyectos-caf---green-climate-fund A simplified impact assessment (ESIA) consistent with the requirements for a category B project is contained in the Environmental and Social Management Framework (ESMF).
Environmental and Social Management Framework (ESMF) (if applicable)	
Date of disclosure on accredited entity's website	2017-08-30
Language(s) of disclosure	Spanish and English translation
Link to disclosure	https://www.caf.com/es/temas/a/ambiente-y-cambio-climatico/proyectos/#itemproyectos-caf---green-climate-fund A management plan (ESMP) consistent with the requirements for a category B project is contained in the Environmental and Social Management Framework (ESMF).
Resettlement Policy Framework (RPF) (if applicable)	
Date of disclosure on accredited entity's website	2017-08-30
Language(s) of disclosure	Spanish and English translation
Link to disclosure	https://www.caf.com/es/temas/a/ambiente-y-cambio-climatico/proyectos/#itemproyectos-caf---green-climate-fund The RPF is integrated in the Environmental and Social Management Framework (ESMF).
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Internal assessment on environmental and social fields based on the studies prepared by the project proponent. The main objective of this assessment is to identify the principal risks associated to the environmental and social potential impacts of the project and to set the guidelines for the ES Management Plan, the Community Stakeholders Engagement Plan and the fulfillment of the required environmental permissions.
Date of disclosure on accredited entity's website	2016-11-02
Language(s) of disclosure	Spanish and English translation
Link to disclosure	https://www.caf.com/es/temas/a/ambiente-y-cambio-climatico/proyectos/#itemproyectos-caf---green-climate-fund

Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Report of the community consultations with stakeholders held august 14 - 18, 2017 and Gap Analysis Document, which identifies the gaps between the performance standards of the International Finance Corporation (IFC) adopted by the Green Climate Fund, the safeguards of CAF – Development bank of Latin America and Argentinean legislation as well as legislation from the Buenos Aires Providence. Both documents are included as annexes to Environmental and Social Management Framework (ESMF).
Date of disclosure on accredited entity's website	2017-08-30
Language(s) of disclosure	Spanish and English translation
Link to disclosure	https://www.caf.com/es/temas/a/ambiente-y-cambio-climatico/proyectos/#itemproyectos-caf---green-climate-fund